

Documentation

All features are part of the WebAnimator Plus program version. A subset is available in the WebAnimator Now program version.



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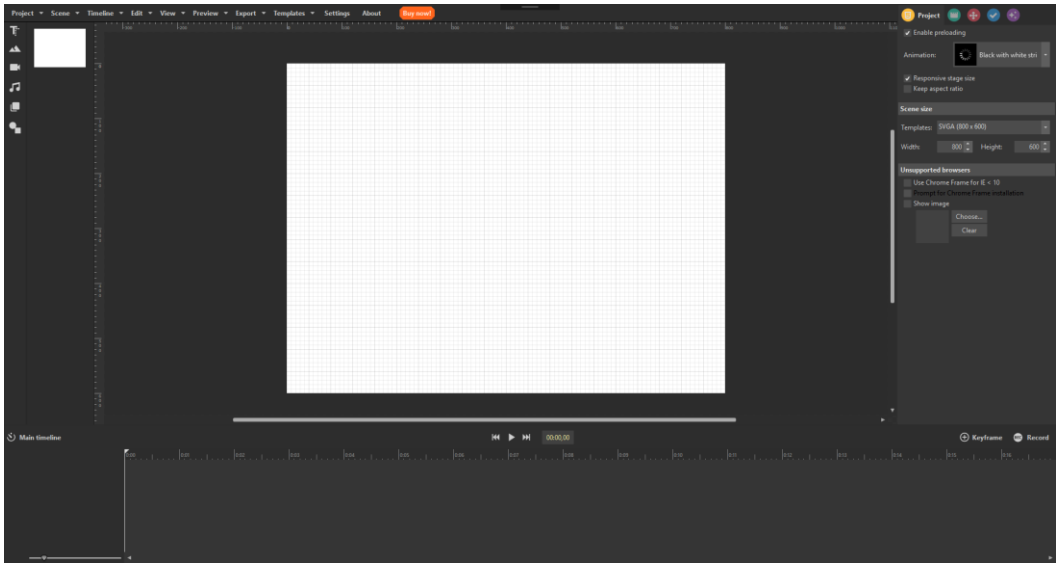
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The Program User Interface

The user interface of WebAnimator is divided into an upper and a lower area.



The lower area, the so-called stage, is your work area for the animations to be created. Below the stage is the timeline which is the main tool and means of orientation for the design of animations. In addition, there are various work palettes for objects, object properties, etc.

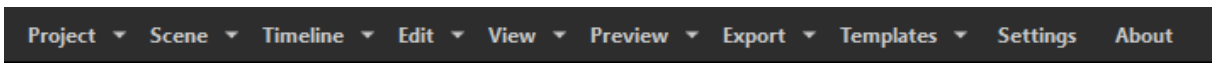
With the modern user interface design, you can simply change between the tabs and work better and more efficient.

The Basic Procedure

Basically, there are two options to create your own animations with WebAnimator. We will explain these options below. For a better understanding, first a brief overview:

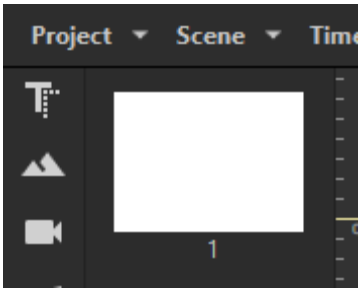
Creating a New Project

When you start the WebAnimator, a new, empty project is opened. The usual file management options *New project…*, *Open project*, *Save project*, *Save project as…*, and *Exit* are available in the main menu under the menu option *Project*.



Creating a Scene

Basically, an animation is always integrated into a scene where simple animations are smoothly added to a scene, while complex ones, such as the ones that change based on interaction with the user, may be divided into several scenes depending on the project.



When you start WebAnimator, you will always see the first scene in the scene area in the upper left part of the window by default. This scene will, of course, be empty yet. The scene area shows a small-size image of what is shown in a larger size on the stage in the center.

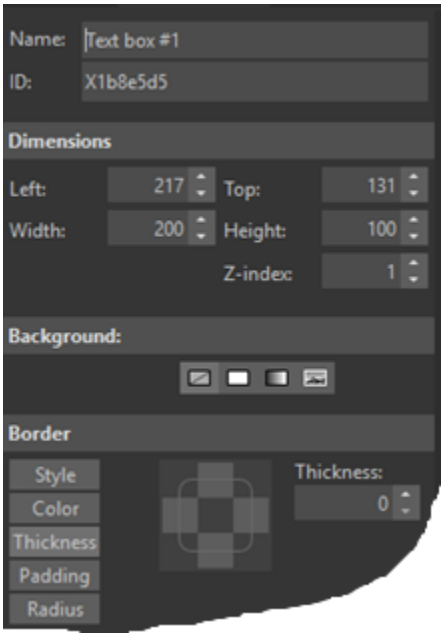
The scenes are numbered consecutively by default. You can, however, number them individually by double-clicking the **1**.

Embedding Objects

You can fill scenes by clicking the item with the left mouse and clicking again the left mouse on the stage or keeping the left mouse button pressed and dragging objects from the objects bar to the stage.

Defining Properties

Afterwards, you can specify the properties for the objects via the context-dependent properties palette.



Depending on object type, this includes, or instance, size, position, color, border, but also font, font color, font size, etc.

You can also define modern CSS3 properties, such as color gradient or rounded corners.

By changing the settings, the changed definitions will be marked with a bold yellow color.

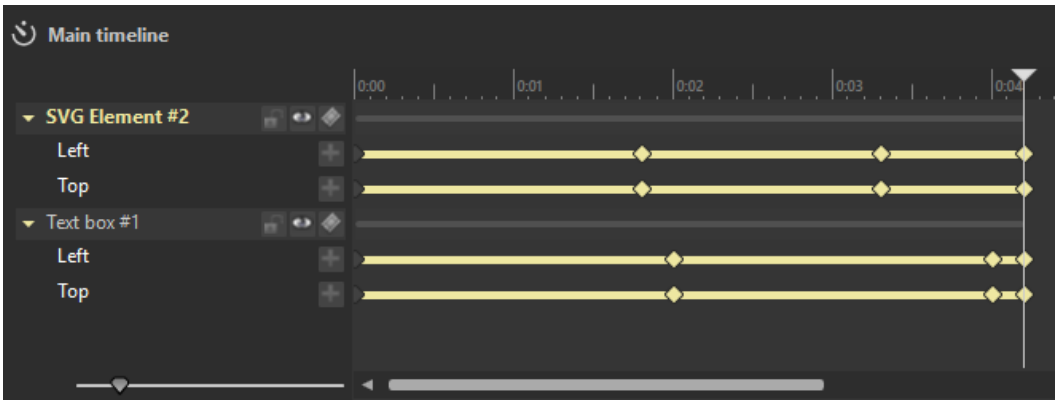
Responsive stage size

With the responsive stage size, you can adjust if you want the finished project in the view to be scaled to the window or if it should be stretched to the window. Keep in mind, that this feature is related to the settings of the scene. With the subitem *Keep aspect ratio* you can define that the objects will be responsive to the window. Otherwise (if the subitem is not checked) it will be relative to the window.

Embedding Animations

Once you have created an object and specified its properties, you can add any number of animation actions of different lengths to the object. For each action, you can specify one or several conditions in which the object has different properties. The animation itself is controlled via so-called keyframes that define the starting point and endpoint of an animation (or of a part of it). Keyframes are inserted into the scene via the timeline.

Keyframes can also be copied and pasted by simply right click on the keyframe line and click on *Copy selected keyframes*. To paste you can simply use **STRG + V** or right click and click on *Paste*.

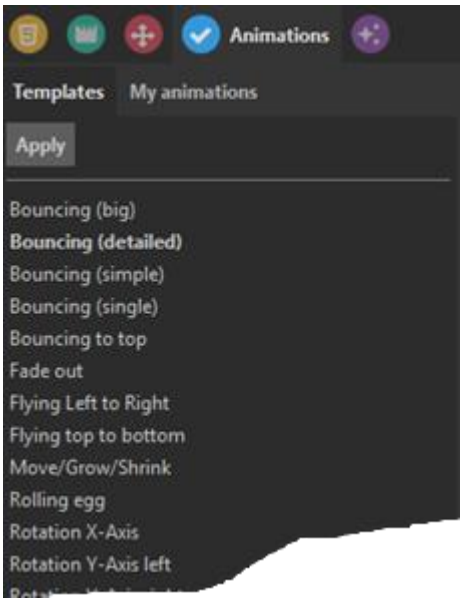


A keyframe is an animation technology term that originated from the production of animated movies. Keyframes provide various basic conditions and rough motion sequences that are then implemented and refined via intermediate images.

If you position an object at the coordinates (10,10) in keyframe 1 and move it to the coordinates (30,30) in keyframe 2, the object will take the shortest path to move to the new position when the scene is played. In the same way, an object can be hidden, a color can be changed from blue to red, and another action can be selected.

Please remember that there are basically two options to create an animation with WebAnimator. On the one hand, the manual process of adding properties in the timeline and setting the keyframes. On the other hand, you can use the recording modus to automatically insert properties and keyframes into the timeline.

Live Animations



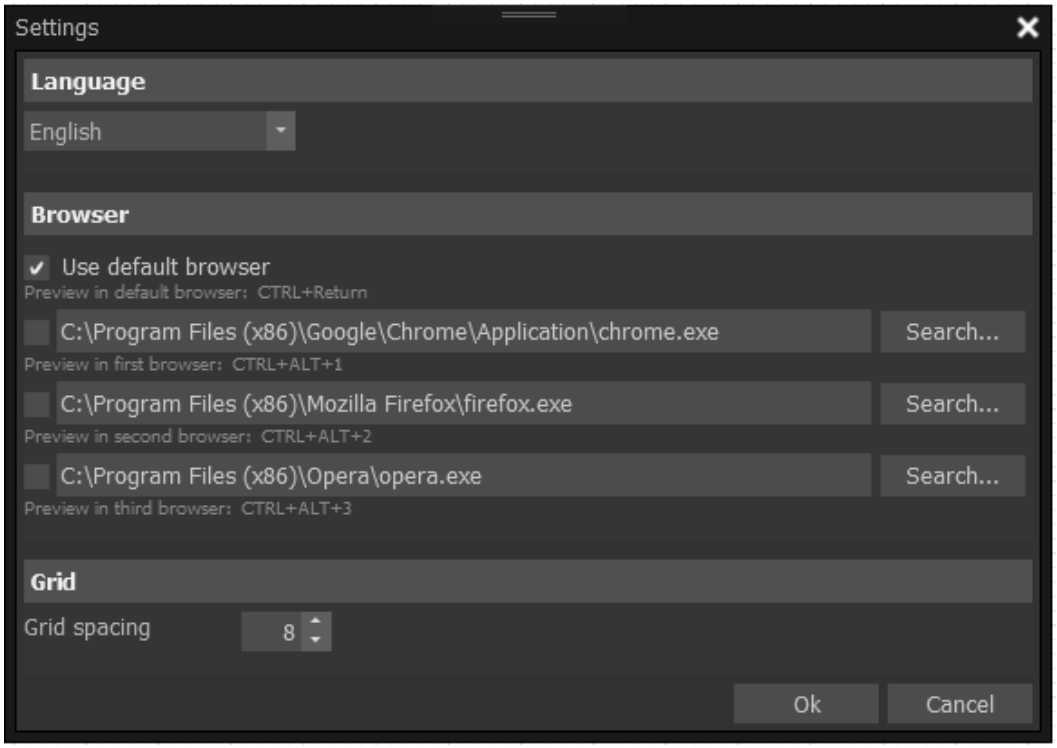
Besides creating animations, yourself you also have the option to realize effects via so-called *Live Animations*.

WebAnimator provides you with various such *Live Animations* which you can apply to all kinds of objects.

You also have the option to save an animation application as your own live animation.

Settings

Via the menu item *Settings*, you can define the browser that should be used to show a preview of the animation. You also can change the language and set up the grid spacing.



Issue Regarding the Support of HTML5 Animations in Various Browsers

When creating animations using HTML5, you always have to consider the fact that not all browsers support his technology. Currently, only the most current versions of the Mozilla (Firefox), Webkit (Chrome, Safari), and Opera browsers as well as Internet Explorer 10 can do so.

Older versions of Internet Explorer, such as the commonly used IE 8 and IE 9, do not include the CSS3 capabilities HTML5 requires.

Unfortunately, even "modern" browsers do not necessarily support all properties in the same way.

For instance, Firefox was not able to display reflection at the time this software was released. The Webkit browsers Chrome and Safari can only display reflections as a static condition, not as an animated change.

Creating and Designing Objects

In the introduction, the steps that ultimately lead to a presentable animation were briefly discussed. In the following sections, we will explain the individual steps in detail and provide appropriate notes, tips, and tricks.

Defining Basic Properties

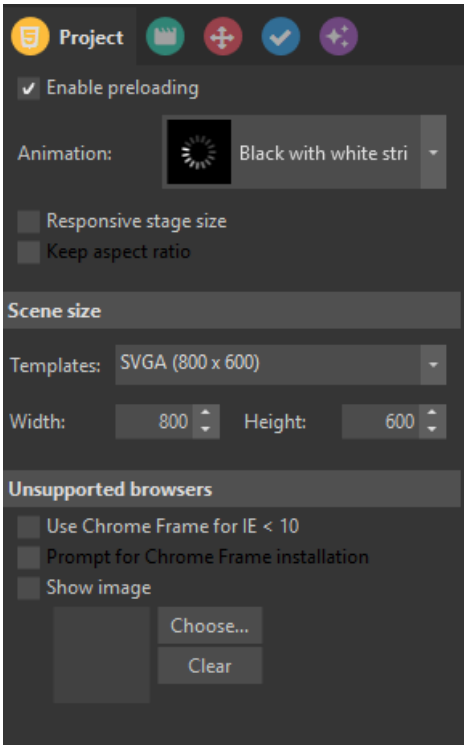
As already briefly described above, you create each animation within a scene. The scene or scenes in turn are embedded into a project.

General project properties

If you have not yet inserted an object into the scene, the context-sensitive Properties palette relates to the current scene and/or the entire project.

First you can decide whether you would like to turn on animation pre-loading. If preloading is activated, your animation will be downloaded completely before it starts playing.

The advantage of using this function is that your animation won’t have to load the necessary resources (e.g. images) while it is playing. Please note: with large animations, pre-loading could cause a delay of the animation start.



To give the user a feedback during the delay, you select an animation to be displayed while the animation is being pre-loaded.

By default, the size of the stage is preset to an SVGA resolution of 800 pixels across by 600 pixels down. However, you can also adjust these values or select one of the many preset values and forms from the Templates drop-down menu. For instance, these templates include templates for animated banners in the skyscraper format up to and including a template of the large full HD format.

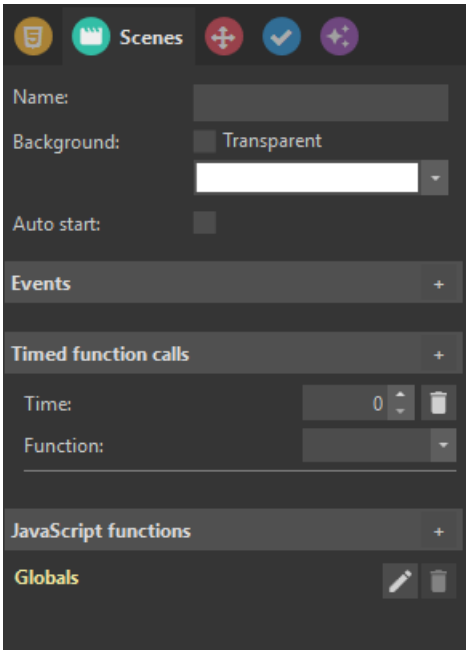
Finally, in the project properties, you can specify what happens when the animation is loaded in a browser which doesn't provide a complete HTML5 support, e.g. Internet Explorer 8 or 9. These browsers wouldn't be able to display the animation; therefore, you should define a static fallback image to be displayed in place of the animation.

You could also try to get your web site visitors to install the free Google Chrome Frame plug-in, which will allow them to display HTML5 content in older browsers www.google.com/chromeframe/). If you activate this function, your animation will be displayed with the help of the plug-in (if installed) or the user will be prompted to install it.

Scene Parameters

By clicking the appropriate tab, you can define the properties of an individual scene. This includes the creation of additional timelines, the naming of a scene, and the definition of various events that occur based on specific user actions.

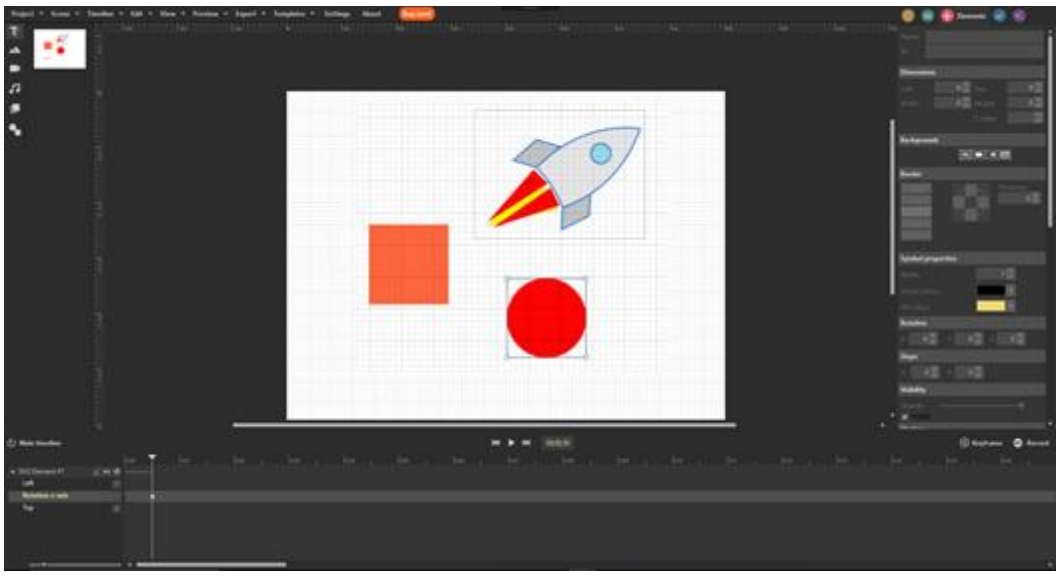
As already mentioned, even complex animations can easily be realized within a scene. We therefore recommend keeping the default settings of the scene for most of the tasks and to only make changes when creating comprehensive, professional animations.



By default, the scene background is white. However, you also have the option to color it as you wish by selecting the desired color from the dropdown menu. If you want to use an image as the background, you simply add it as an image object in the size of the entire stage.

Creating an Object

Objects are the essential components of your animation. These objects can be geometric forms, such as circles or rectangles, as well as text boxes, images, and even videos and sounds.

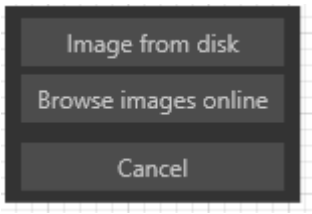


As already briefly described above, you can insert objects via the *Objects* bar. To do so, click the button of the desired object and drag it to the scene while keeping the left mouse button pressed.

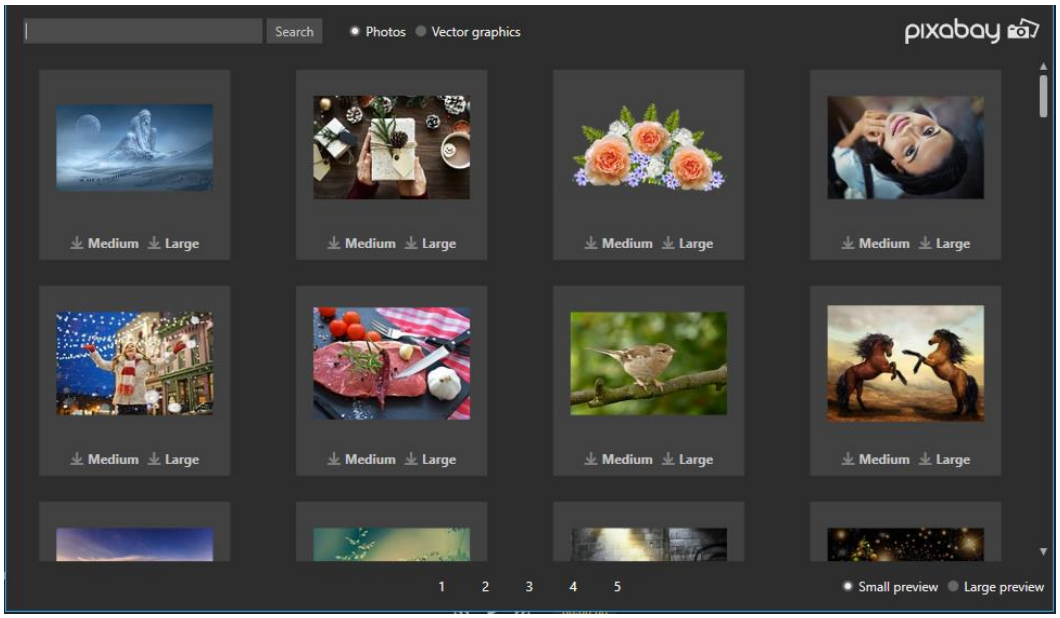
You then get either a standard object based on internal templates (in case of forms, text boxes, lines, and buttons) or you are prompted to select a source file (in case of images, video and audio files).

The objects you added are nothing more than placeholders at this time. You must add properties for them and define the actual animations. You can, of course, also rename them by double-clicking the default name in the left part of the timeline and entering the desired name.

Importing Images



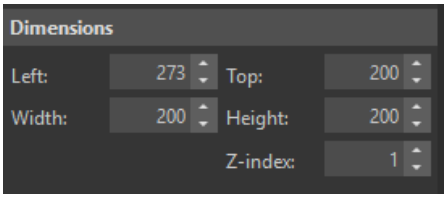
Images are one of the essential components of your animation. You have the possibility to put images to your project from your computer and Browse images online from pixabay.com. Supported Images are Bitmaps, SVGs, and any other kind of images like .jpg, .png etc. By clicking the image icon on the top left, there will appear a window where you can decide if you want an internal or external image.



Assigning Object Properties

The object(s) added form the basis for further editing. To do so, click the desired object in order to select it. Keep in mind that you now have access to context-sensitive commands via the *Properties* palette. These commands change depending on the type of the selected object.

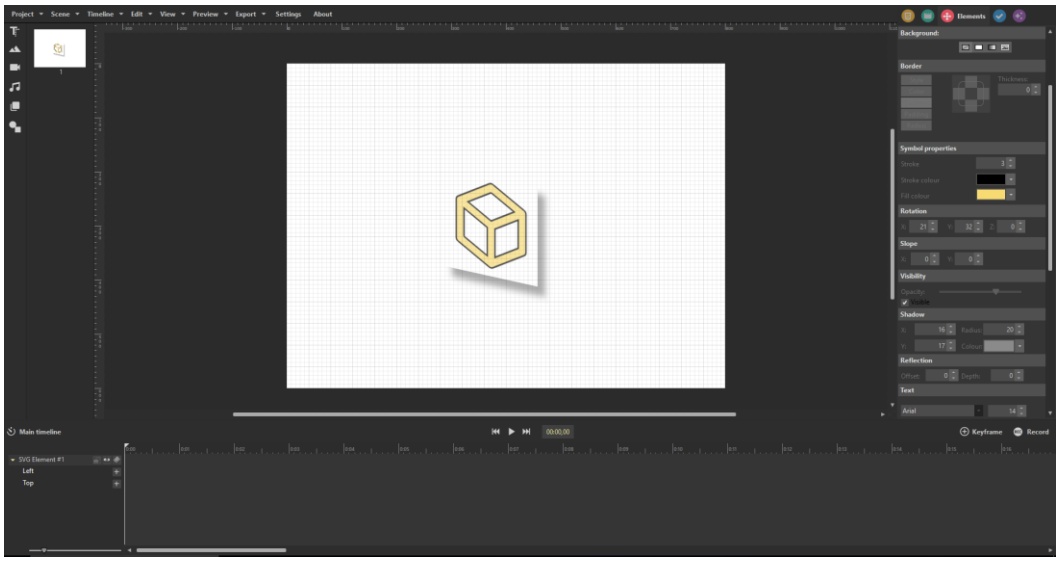
When you drag an object to the stage, it always has the two basic properties *Left* and *Top* which define its position on the stage. The *Project* and *Scene* tabs which you have already learned about on previous pages remain the same.



Now, you can, for instance, use the tab *Dimensions* to change the size of the element. Of course, you can grab the anchor points to drag the object to the desired size as well, but the *Properties* palette lets you specify the exact size of the object as well as its position on the stage in pixels.

Now, click the *Element* tab in the palette. This gives you access to comprehensive modification options that you can use to add a variety of properties to the object. You can, for instance, add a color gradient to the object via the Background item or select an image as the color of the object.

In the *Rotation* and *Slope* sections, you can set the appropriate properties, and in the *Border* section, you can select a frame with various properties.



The *Visibility* setting gives you different manipulation options for the fading in and fading-out of objects.

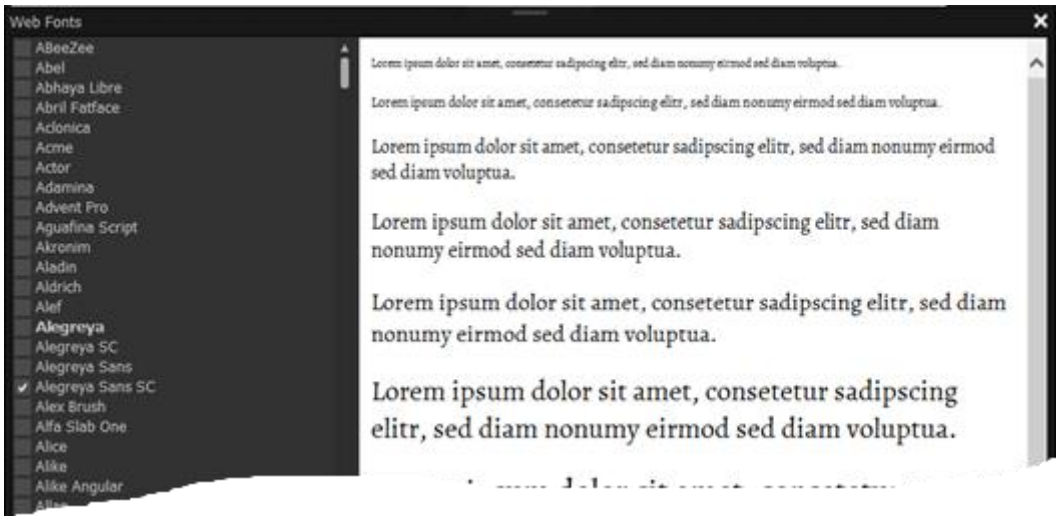
Lastly, you can also add the details of a shadow or reflection to the object. Once you have added all the necessary properties to the object(s), you can start with the animation.

Special Case: Text Objects Inheriting HTML and Google Web Fonts Directly

If you use text boxes as objects, you can edit the standard specifications, which you probably already know from other text editors, such as font, font size, font color, etc., via the *Text* tab of the *Properties* palette.

Beside regular fonts like Arial, Times New Roman, etc., WebAnimator allows you to use Google Web Fonts. From the tab *Text* of the properties window, select the last entry *Choose Fonts*, to get a direct access to Google Web Fonts and apply the online fonts to your text.

The formatting is shown directly on the stage, like in the browser preview.



However, you can also edit the HTML code of the text box, the so-called inner HTML, directly. To do so, click the text object and then click the box on the right bottom of the page. In the box

that opens, you can enter the inner HTML code directly which will format the text of the object accordingly. You may also copy HTML code from other sources into this box to use it in the text box.

Adding Animations to Objects

Whether you first place all the objects required for the animation onto the stage, assign the basic properties to them, and then start to place the animations, or whether you first animate the objects and then change their properties is unimportant and only a matter of your personal preference.

Just experiment and see which procedure works better for you and your project.

Manual Procedure versus Recording Mode

Assigning properties alone does not turn objects into animations. Animation basically means that object properties take on certain predefined values at certain predefined points in time.

These values and their respective points in time are represented in the timeline which is one of the most important tools besides the stage.

The interaction between the stage and the timeline basically lets you create animations in WebAnimator in two different ways:

First, you can create changes manually where you can make very detailed and precise edits.

Second, you can also use the so-called recording mode. This mode is somewhat easier to use but does not let you make edits as detailed as the manual mode without post-processing.

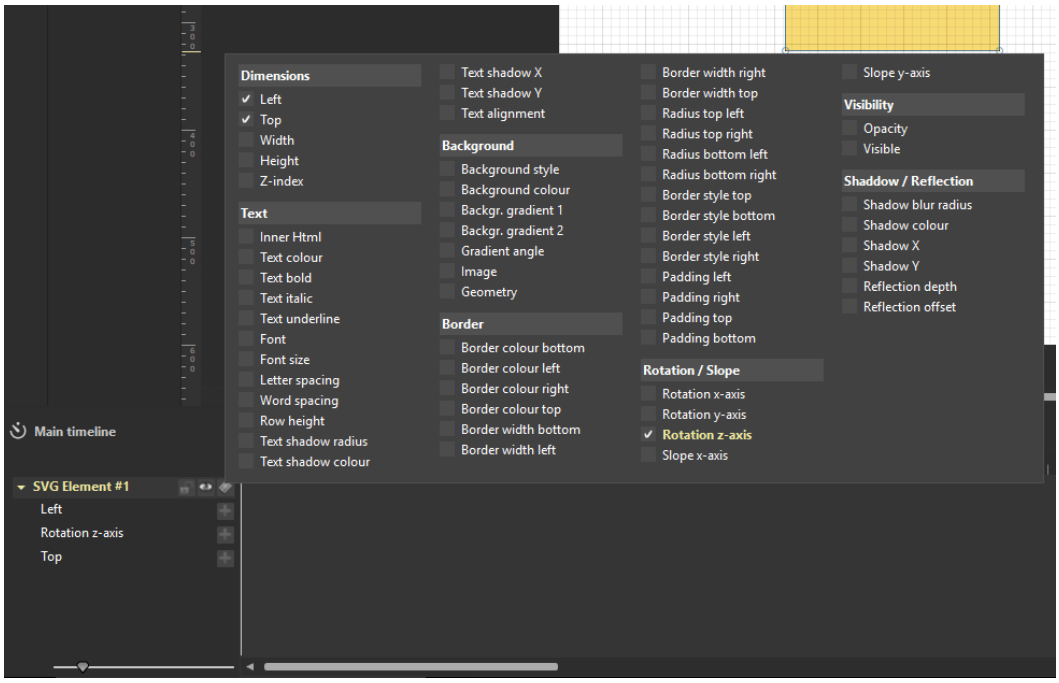
It is hard to say which is preferable as it depends greatly on your experience and requirements. With practice, you will probably use both modes for different occasions.

Animations using Manual Mode

In manual mode, you add the properties that you want to animate to the objects by looking for the object in the timeline below.

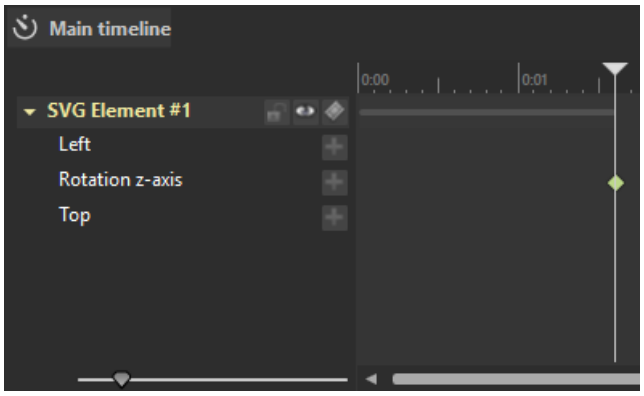


Next to the name of the object, there are also small buttons. The button in the center is used to show and hide the object on the stage. The left button is used to lock an object on the stage. When the right button is clicked once, the list of object properties opens.



You select an object property here by setting a check mark, for instance, for the *Rotation z-axis*. This property is then added to the timeline. Now select the newly added property and move the cursor to the position in the timeline where the first keyframe is to be created. This is where your animation will start.

Then, either click the small plus sign to the right of the property description or click the Keyframe button above the timeline to the right.

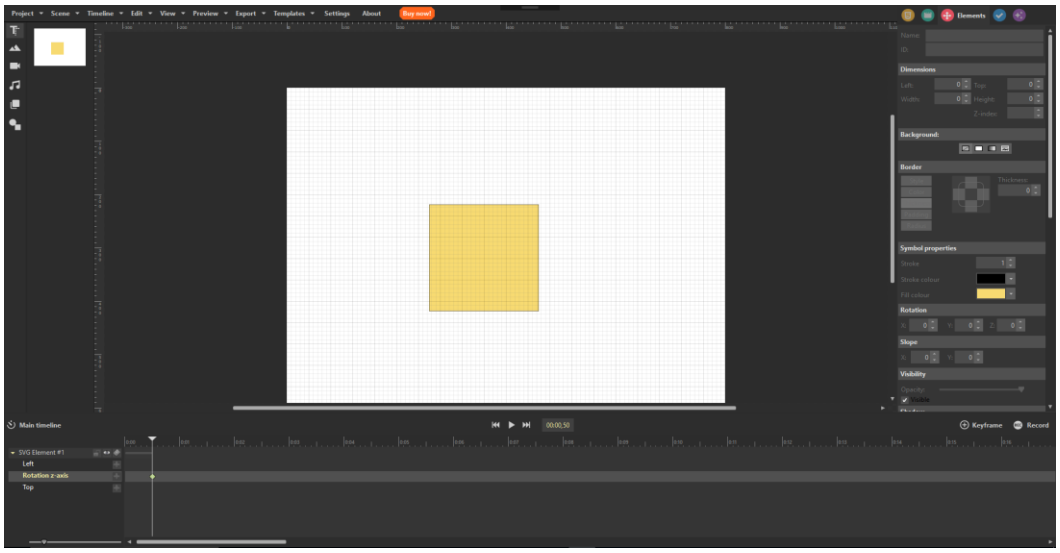


Now specify the value of the newly created keyframe. To do so, go to the *Properties* palette to the *Element* tab (the respective element must be selected) and under *Rotation* change the value for z (this is the property that you manually added to the object in the timeline of our example above). If the original value is exactly what you want, there is, of course, nothing to be changed.

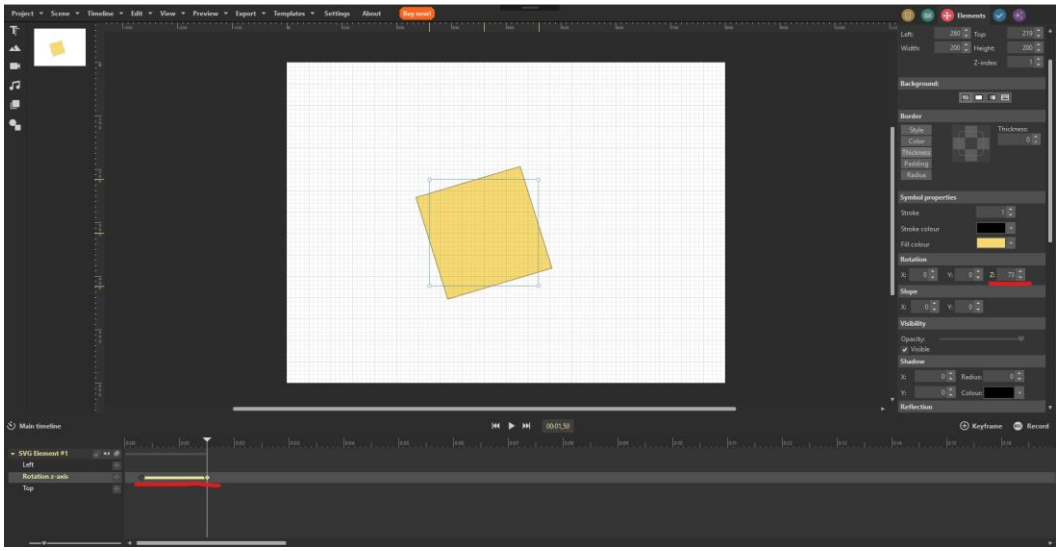
Now, position the timeline cursor at the desired location and create the second keyframe there using the plus sign.

Once again, change the value of the object property via the appropriate window.

You can see right away that the intermediate steps between the two keyframes are generated automatically.

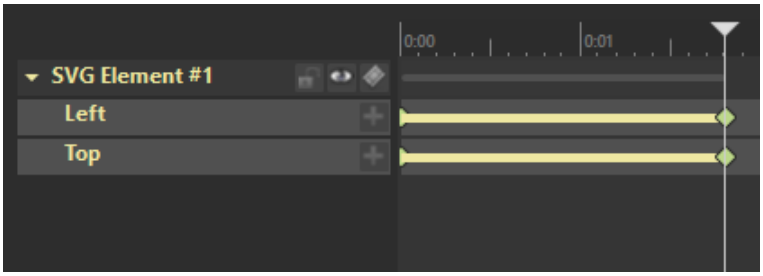


You can now go to the beginning of the animation, i.e. the first keyframe, or to the beginning of the timeline and use the control elements above the timeline to run the animation. Congratulations, you have just created your first animation using WebAnimator.



Selecting Several Properties at the same Time

Some object properties are affected by not just one but by two or more parameters. For instance, the position of an object on the stage is defined by the two properties *Left* and *Top*.

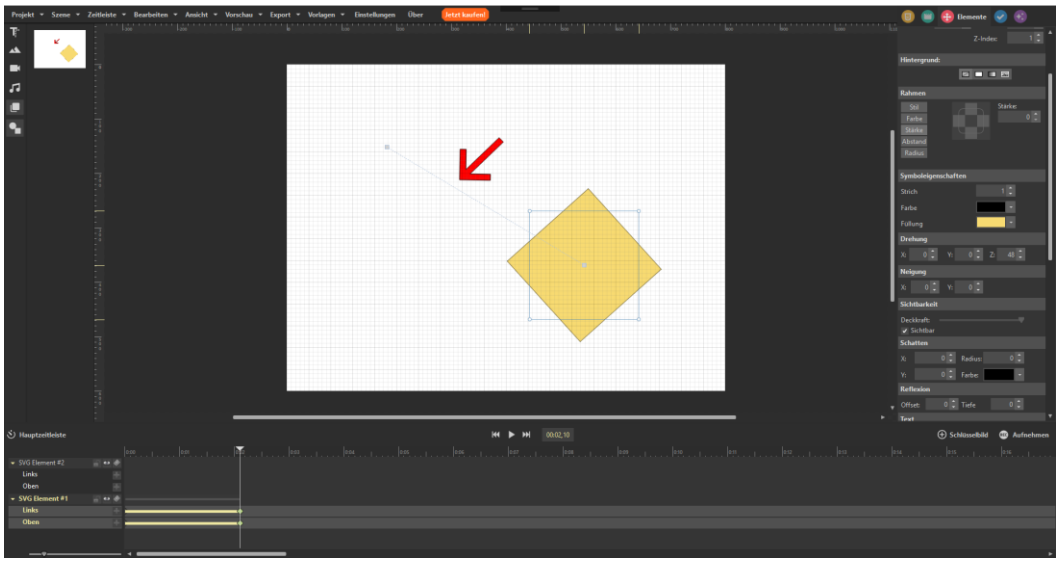


You can change both properties at the same time if you select them both using a selection frame. You can then create keyframes for two or more properties via the Keyframe button above the timeline to the right that you already know.

Then, proceed as follows: Add the keyframes at a different position in the timeline and change the properties.

To keep with the example of the position of an object, you could now drag it across the stage to a different position or change the object properties (in this case via the *Dimensions* tab, the properties *Left* and *Top*). The intermediate steps are once again calculated automatically and added.

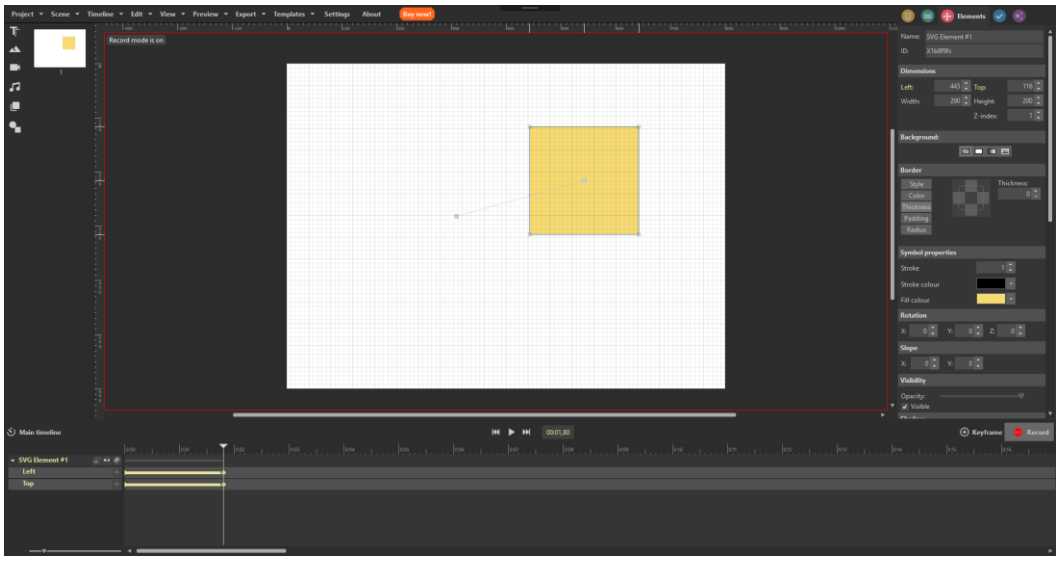
Please note that the path of the object, the animation path, is shown as a guide that will, of course, not be visible in the finished animation.



Recording Mode for Animations

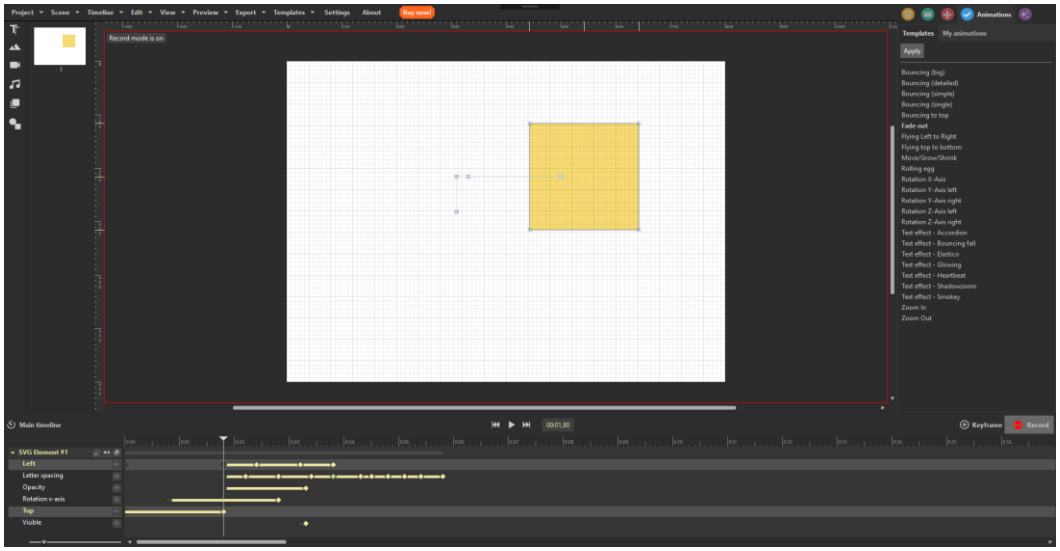
Besides the manual mode which gives you access to the most complex control options, you also have the option to use the automatic recording mode. Select the object as you have seen above and click the *Record* button above the timeline to the left.

For instance, if you now set the timeline cursor to the desired position and move the object, you will see that the keyframes and intermediate steps are created automatically.



Now, move the cursor along the timeline and move the object further. The keyframes are always automatically generated at the point where the cursor is located.

If you change a property that is not yet in the timeline, it is automatically added. You do not have to select it specifically from the list of properties.

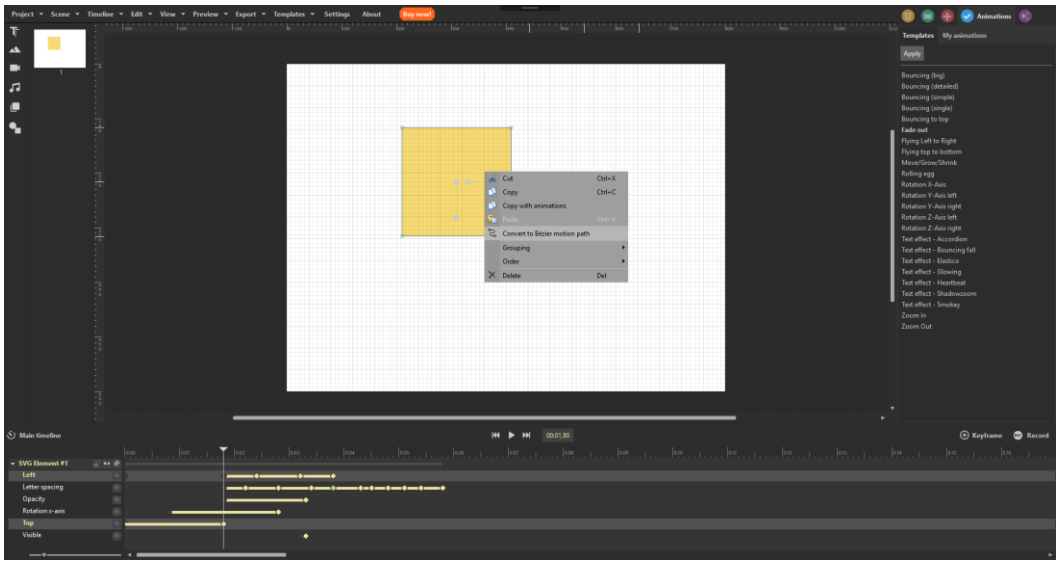


In this way, you can design animations in easy steps. Once you have constructed all steps according to your wishes, click Record again to exit recording mode.

Of course, you can post-process any animations that you created via the recording mode by editing existing keyframes or adding new ones manually. You can also use a combination of both modes which will probably become the rule in your work.

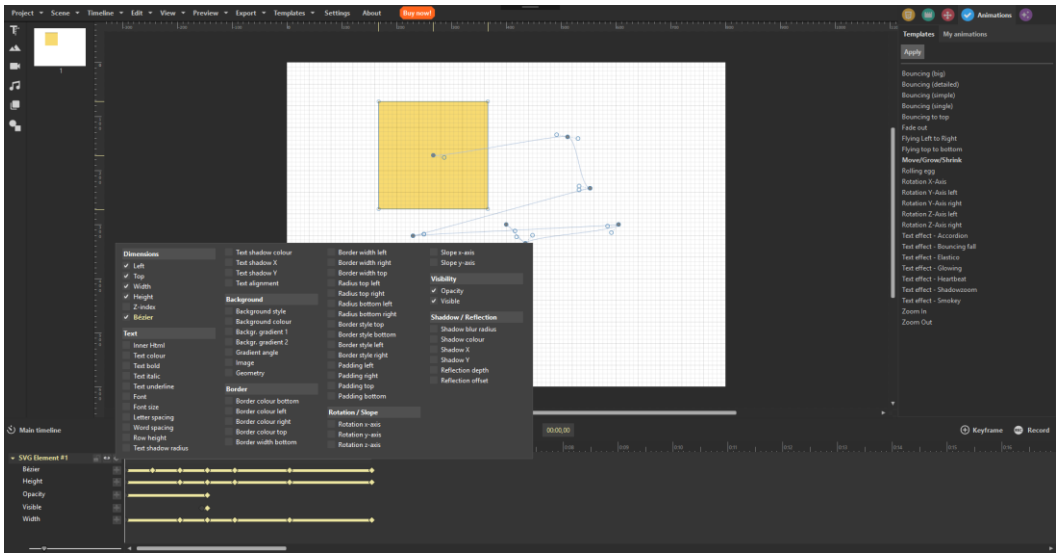
Creating Bézier Animations

Besides polygonal or straight animations, you can also easily create arched, curved, or circular animation paths. You have two options to create such Bézier animations: First, create a regular animation. Then, click the object and select the option *Convert to Bézier motion path* from the context menu.



If necessary, you can then adjust this Bézier motion path to make it even rounder or smoother, for instance.

However, you can also specify at the beginning that an animation is to contain a Bézier path by choosing the *Bézier animation* via the respective selection in the timeline and creating the keyframes as shown above.



Using Videos and Audio Files in Animations

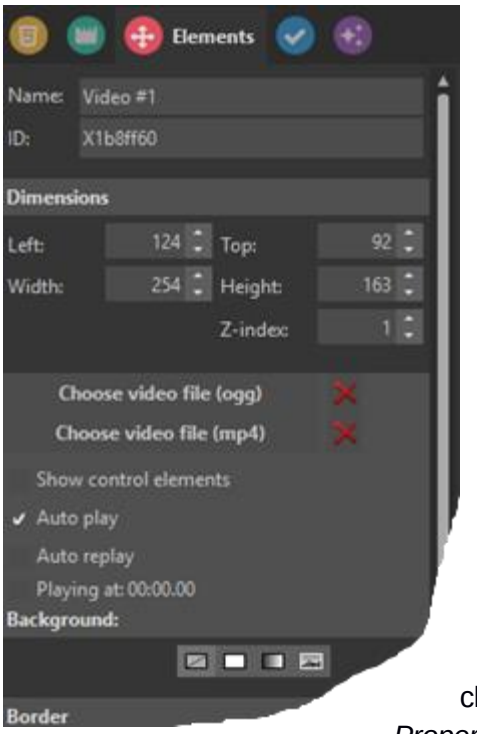
Besides objects, such as images and geometric forms, you can also use movies or audio files as animation objects. By now you can also add YouTube videos to the stage by entering the URL of the YouTube video in the box. Doing so, however, creates a problem that is due to the specifications of HTML5 and the way HTML5 is used in various browsers.

By means of HTML5, web developers can integrate video and audio files directly into their web pages, and visitors can view these files without the need for any additional plug-ins. A few operators in special tags, such as specifying the source, adjusting the size, displaying control elements, etc., are enough to be able to play videos and sounds. The problem is that there is currently disagreement as to which codecs are to be used in these elements specifically.

Even though the respective elements are clearly defined in the HTML5 specification, the specification does not contain any information regarding the codec support to be used. This results in the two most widely used browsers, Internet Explorer and Mozilla Firefox, not using the same format, for instance.

While Firefox and Opera expect the free video compression format OGV, for instance, Internet Explorer and Apple's Safari browser insist on MP4 with H.264 codec. Only Google's web browser Chrome supports both formats.

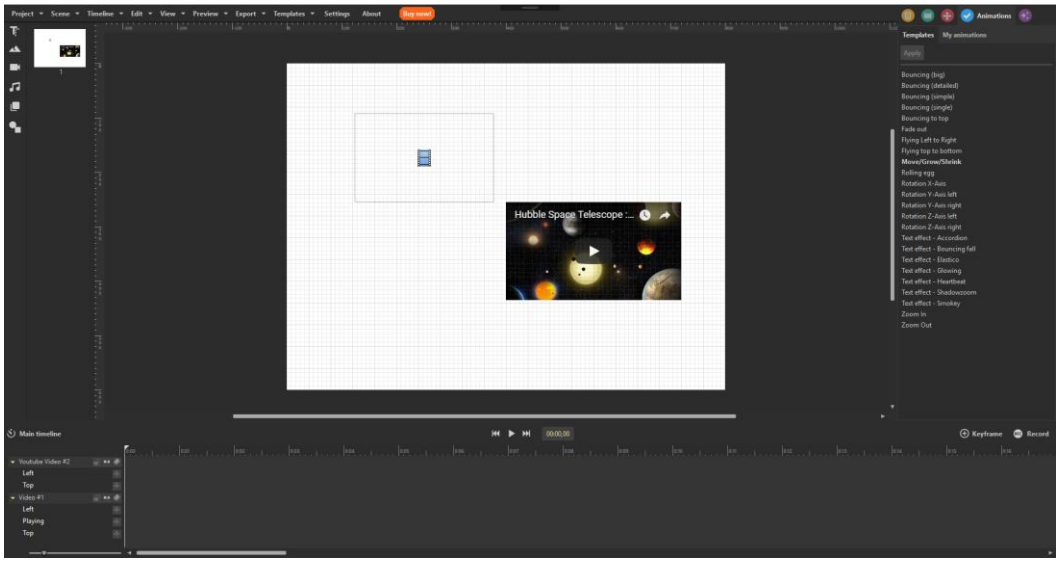
The consequences are obvious: If your animation is viewed on an iPad using Safari and you only provide OGV, the user won't see a video. The same is true for MP4 and Firefox. The situation is very similar in case of audio files, but here it concerns the OGG, MP3, and WAV formats.



Even though, right now, it looks like the development of additional video codecs could put an end to this "format war", you do not have any other choice in the meantime but to convert video and audio files to both formats and to make both formats available to the user. Depending on the browser used, the specific format is then selected automatically. Thus, in order to provide video or audio files in an equally convenient way, you must create them in all the formats and give them the same name for practical reasons, not out of necessity, so that they only differ in their extension.

Then, drag a video or audio object onto the stage in the same way you would geometric forms or images. Then, you must select a source file that WebAnimator is to integrate.

If you select the object now, you can see the option to choose the different file formats individually in the *Object Properties* palette. Click the respective button and select the respective file. Here, you can also specify whether the files are to be started automatically within the animation, whether they are to be replayed at the end, or whether the control elements are to be displayed automatically.



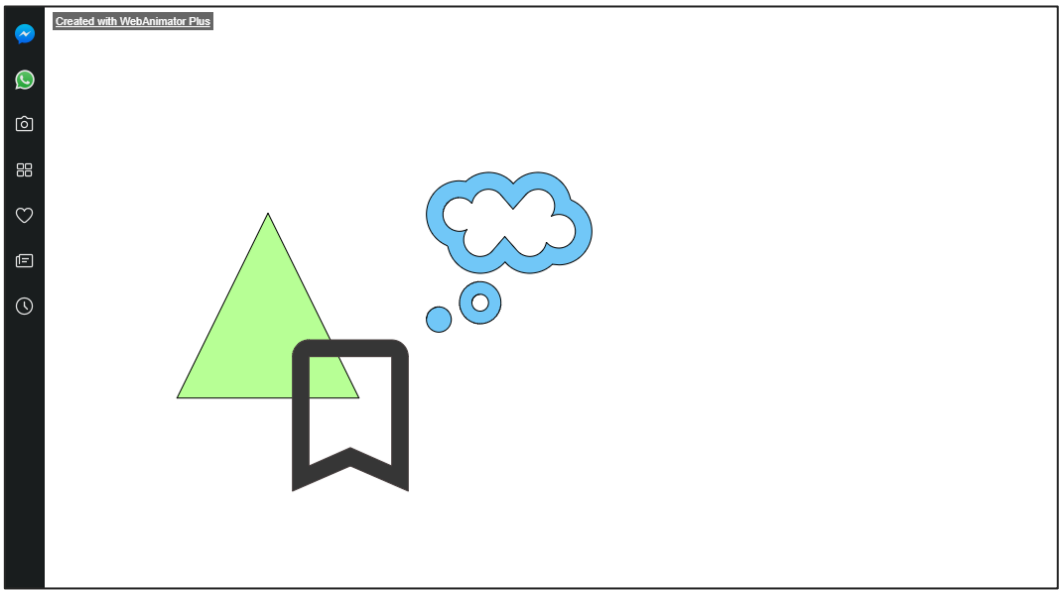
Testing Animations Using Preview

You have already heard about the preview function in the section *Settings* above. This preview function lets you test the actual animation in the assigned browser during project development.

To do so, you can either select *Preview* from the menu bar or select a specific browser via a keyboard shortcut:

- Preview in standard browser: **[Ctrl]**+**[Enter]**
- Preview in the first browser specified: **[Ctrl]**+**[Alt]**+**[1]**
- Preview in the second browser specified: **[Ctrl]**+**[Alt]**+**[2]**

- Preview in the third browser specified: **[Ctrl]**+**[Alt]**+**[3]**



Post-Processing in the Timeline

If you are not quite happy with an animation, e.g. if it is running too fast or too slowly, you can post-process it directly in the timeline.

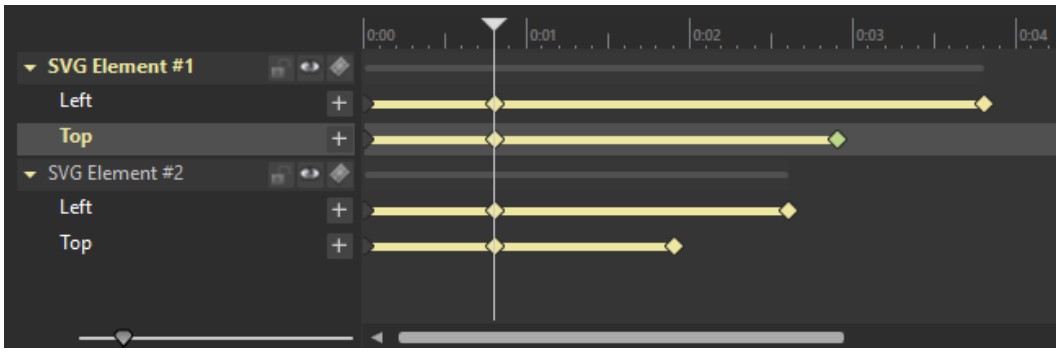
Changing the Duration of an Animation

The duration of an animation can very easily be adjusted by clicking on a keyframe and dragging it while keeping the left mouse button pressed to make it longer or shorter.

In order to change the point in time of concurrent keyframes, you can select them using a selection frame or by clicking them one after the other while pressing the **[Ctrl]** key and then drag them to the desired position. You can, for instance, grab all end points and move them to a different point in time.

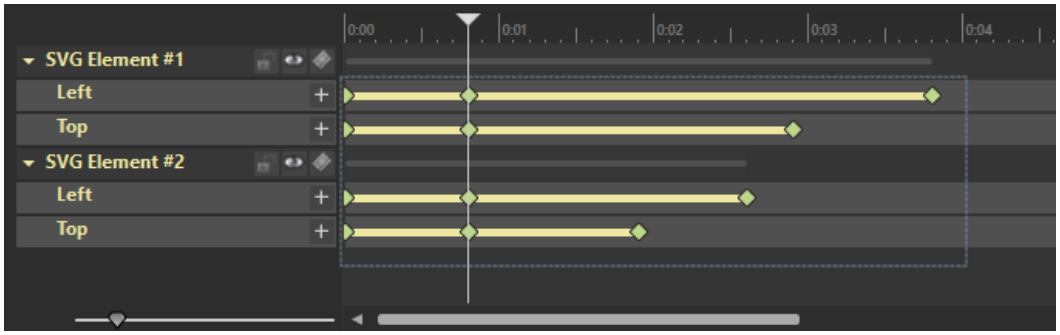
In doing so, the structure of the keyframes remains the same.

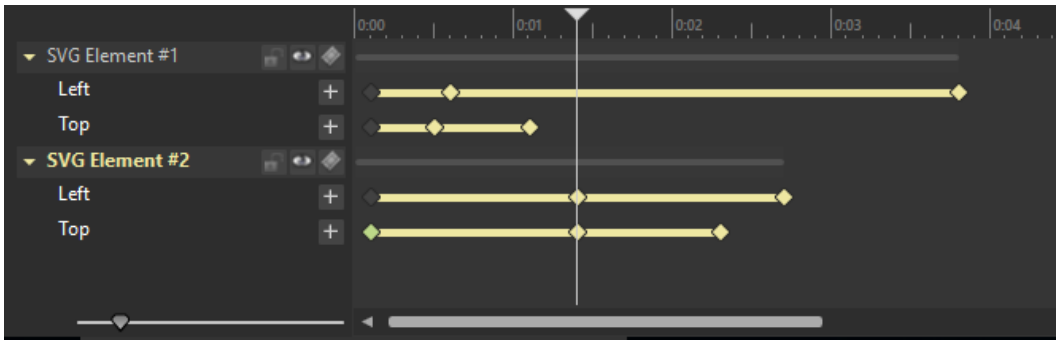




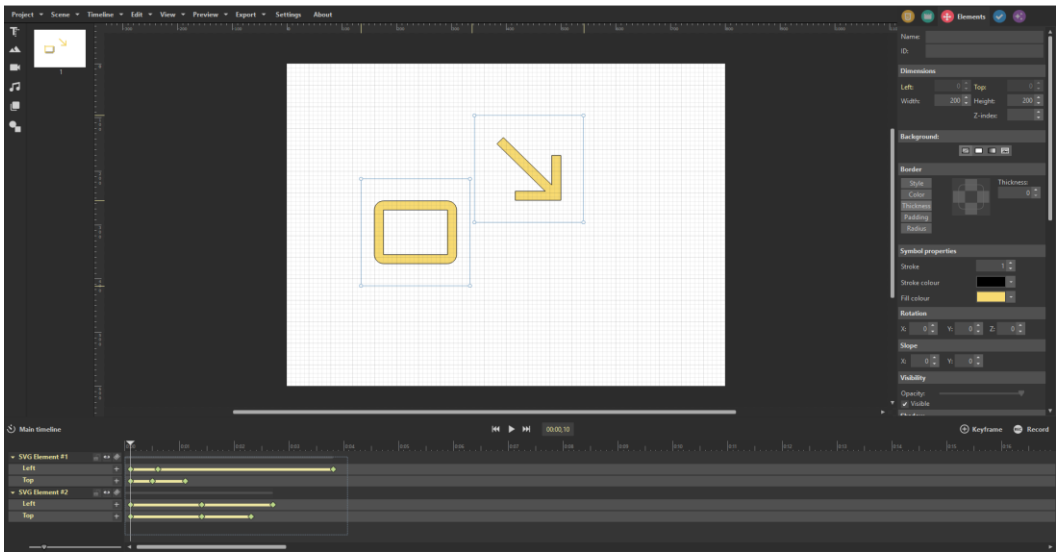
Changing the Start/End Time of an Animation

If you, however, select the entire structure, you can move it as well and thus change the starting point or end point for the entire animation.



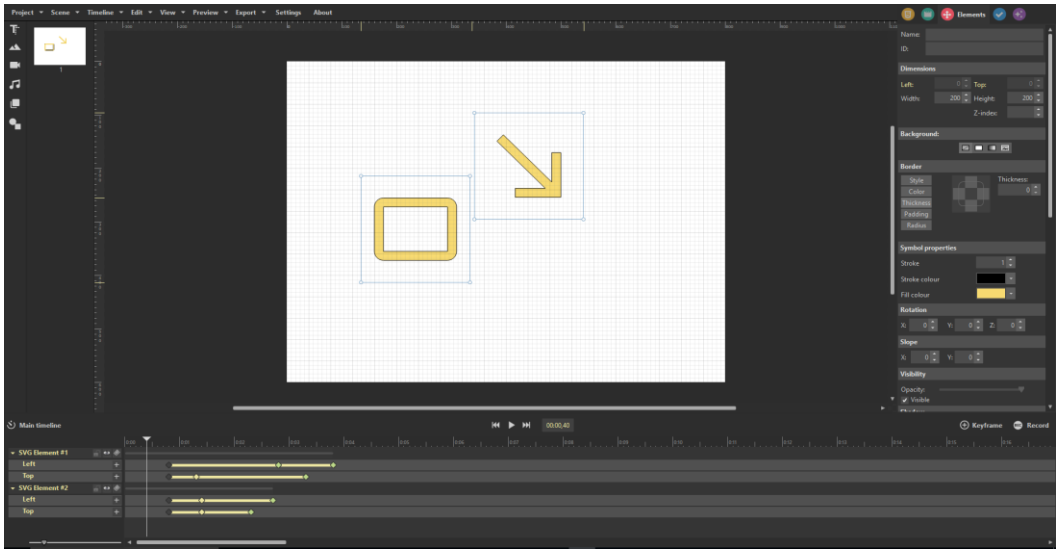


You can, of course, also select and move individual keyframe paths in the timeline so that they run consecutively instead of concurrently.

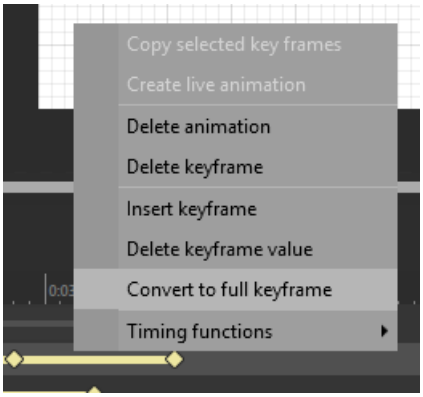


In doing so, please remember one special case in connection with positioning objects: If the timeline cursor is directly on one of the keyframes for the properties *Left* and/or *Top* and you move the object, you change the animation path and, in this case, also the value of the keyframe.

If the timeline cursor is not on one of the keyframes of the properties *Left* and/or *Top* and you move the object to a point outside the stage, you change the position of the entire animation on the stage.



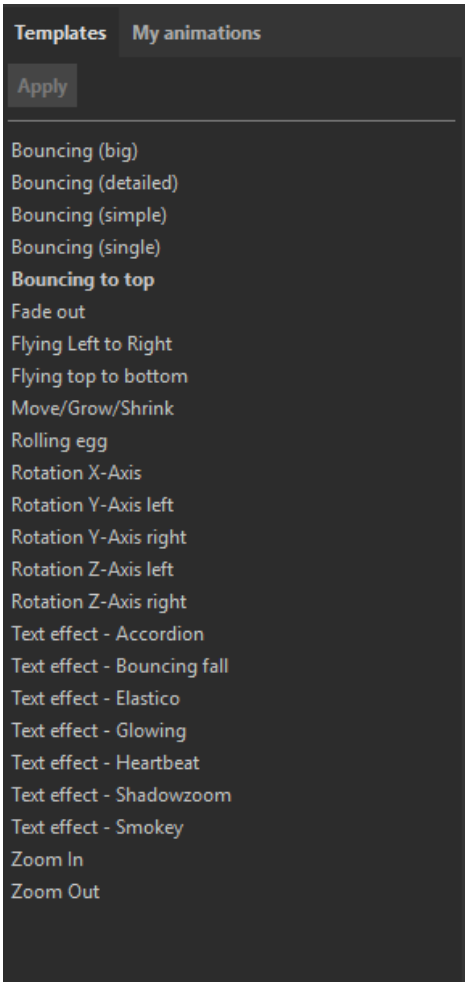
Just as easily as you can move and reposition keyframes and/or their paths, you can also change the connection between keyframes and their values. To do so, right-click a keyframe. The context menu gives you various options.



For instance, you can delete the keyframe or insert a new one. The new keyframe will be created at that point in the timeline where the time indicator is located. You can, however, also just delete the value of the keyframe and then convert it to a keyframe with a value.

Applying Live Animations

A special feature of WebAnimator are the *Live Animations*. They are located in the respective tab of the library at the right edge of the screen *Live Animations* basically make your work with projects easier. By using prepared effects and applying them to your objects directly, you can reduce the amount of work a project requires.



In this context, "live animations" refers to everything that you could also create using the properties in the timeline. This includes the moving of objects between two points, changing the color or size, editing texts, etc. To use a *Live Animations*, you simply select an appropriate one from the list and drag it to the desired object while keeping the left mouse button pressed. In this way, it is automatically applied to this object.

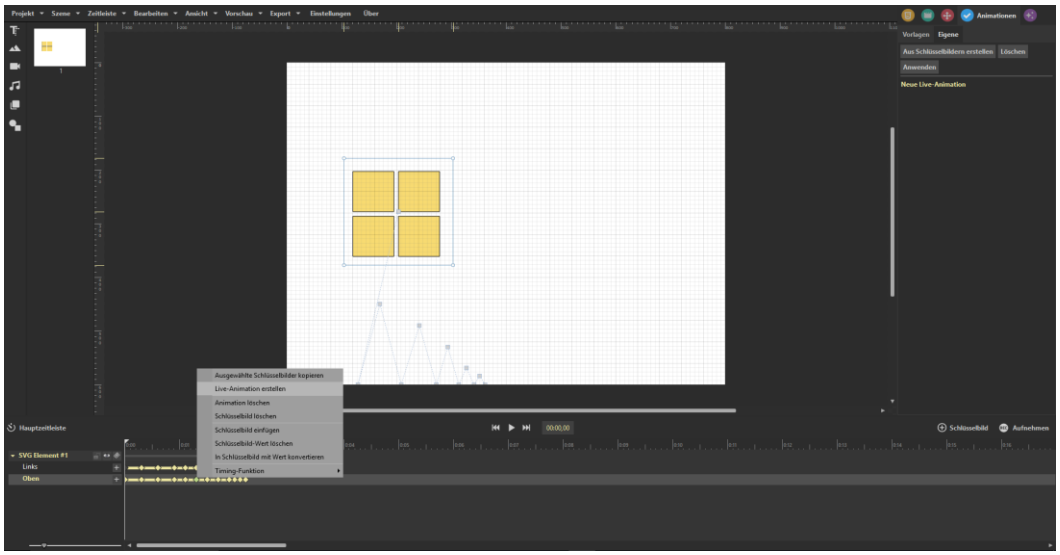
Do however remember that you cannot apply certain text effects to a geometric object, for instance. Of course, you can still edit the *Live Animations* you applied to the object afterwards if some of the details are not exactly what you want.

If desired, you can also apply multiple *Live Animations* to the same object and thus combine the individual effects.

Creating Your Own Animations

Besides using the animations provided, you also have the option to save exceptional animations that you have created yourself for later use. To do so, click in the timeline and select all of the frames of the animation using a selection frame. You can then right-click and select the option *Live Animations* from the context menu.

You can however also go to the *My animations* area of the *Live Animations* tab in the library and click *Create from keyframe* to save the animation.



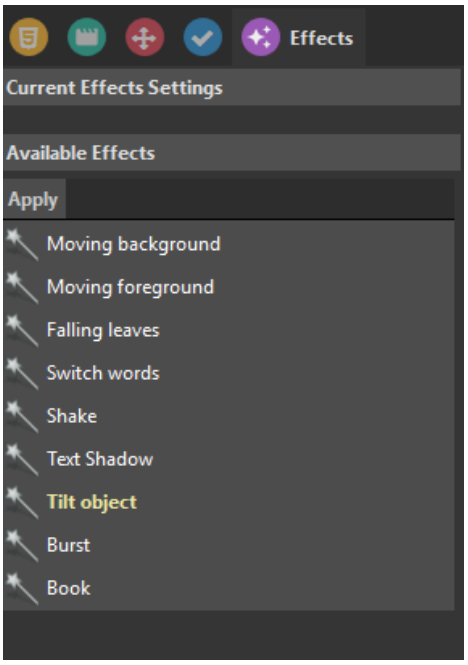
In this way, you save your animation in the live animations library and can apply it via drag and drop or clicking to other objects. Of course, you can also rename your *Live Animations* by double-clicking the title and entering an appropriate name.

If you want to delete an animation, you can easily do so by right-clicking it in the library and selecting the appropriate option from the context menu or by selecting it and clicking the *Delete* button.

Using Effects in Animations

Contrary to *Live Animations*, the effects located in the next tab of the *Library* are not based on keyframes, i.e. they cannot be post-processed or adjusted in the timeline.

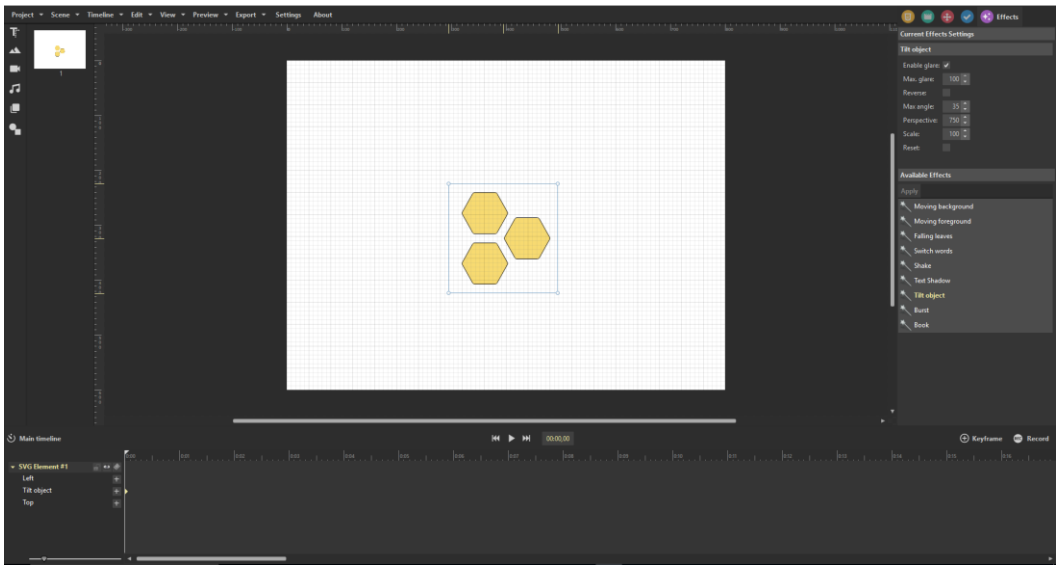
Rather, these are special effects that were implemented using JavaScript.



In order to use an effect, you simply drag it – just like *Live Animations* – via drag and drop or by clicking onto the stage or, in case you select the Switch words effect, onto a text object. Afterwards, you will see the entry of the effect in the timeline, but at the same time, a configuration window will open in which you can define the general behavior of the respective effect, in our example *Tilt object*.

Now change for each Effect the settings which you like most. Each effect has a different settings box, so that some effects may not work with specific types of objects.

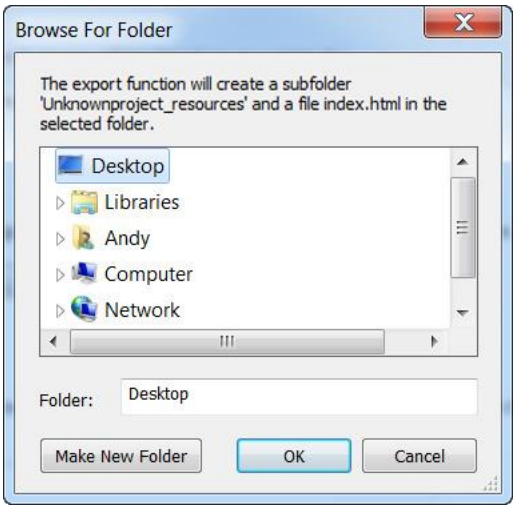
With checkboxes you can set your settings and test them after in your browser.



Remember that the effects cannot be seen on the stage in the program itself but only in the preview in the browser.

Exporting an Animation

In this section we will show you, how you can export an animation as a complete HTML page including all embedded files and how you upload them to your web server to integrate the animation in your website. Click on *HTML5 Export* in the menu bar and choose a location to store the exported animation.



You can either select an existing folder or create a new one. Then, click *OK* to start the export.

WebAnimator will scale down the embedded images (if you used any) to the maximum required size and create the pre-defined folder structure. In the chosen storage folder you will find the generated HTML file containing the animation (e.g. *myproject.html*) as well as a separate resource folder (e.g. *myproject_resources*) holding the animation assets, such as images, scripts, etc.

Embedding an Exported Animation

You can upload the exported animation to a web server and use it as a separate website or integrate it into an existing website. To use the animation as a separate website, upload *myproject.html* and the corresponding resource folder to your web space using a FTP program.

Example: Let's assume your URL is www.mydomain.xy. You have created a subfolder animation in the root directory of your server with the help of a FTP program and uploaded the HTML file myproject.html and the resource folder. If you load the *URL www.mydomain.xy/animation/myproject.html* in your browser, the animation will be displayed.

To embed your animation into an existing website, open the file *myproject.html* with a text editor of your choice (e.g. Notepad++, PSPad or just the standard Windows text editor).

Copy the three code lines (the complete Div element) and paste them in the page where the animation should be displayed:

```
<div id="myprojectStage" class="a5animatorscript">
      <script type="text/javascript" charset="utf-8"
      src="myproject_resources/WA_SCRIPT.js"></script>
</div>
```

After that, copy the following line:

```
<script type="text/javascript" charset="utf-8"
src="myproject_resources/jquery-1.7.min.js"></script>
```

and paste it into the Head section of the web page.

To finish the integration, upload the resource folder (***myproject***_*resources*) via FTP to your web server (the folder should be put in the directory containing the web page in which you would like to integrate your animation).

You can embed more than one animation per web page. Select the three code lines (Div element) of the second animation and paste them into the desired page. The line for the Head section needn't be pasted again. Don't forget to transfer the resource folder of the second animation to your web space and you are done. Repeat these steps for all further animations.

If you are using Chrome Frame in your animation, the embedding code should be extended a little. Place the following additional code segment in the *Head* section:

```
<meta http-equiv="X-UA-Compatible" content="chrome=1">
<script type="text/javascript"
src="http://ajax.googleapis.com/ajax/libs/chromeframe/
1/CFInstall.min.js"></script>
```

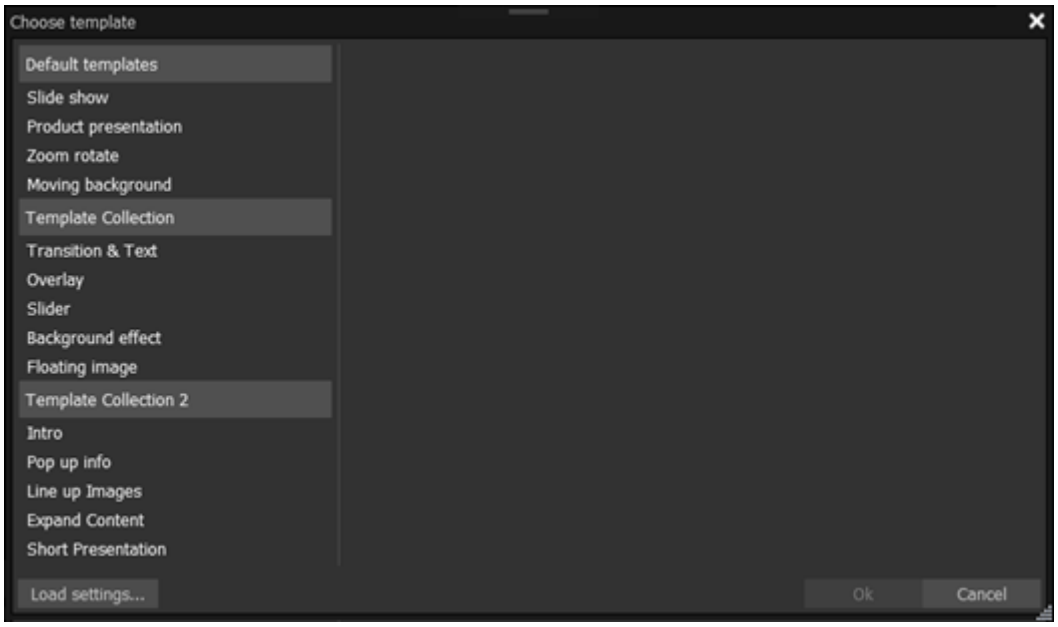
Add the following script in the Body section of the page below the Div element(s):

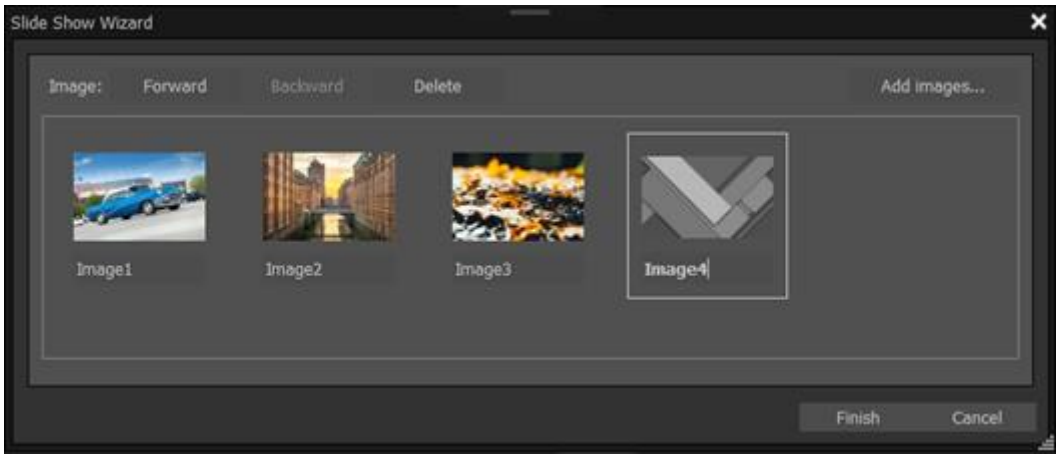
```
<script type="text/javascript">
/* <![CDATA[ */
if ($.browser.msie && $.browser.version < 10)
{CFInstall.check({mode: "overlay"});}
/* ]]> */
</script>
```

Alternatively, you can copy the complete code from the exported HTML file into your web page.

Quick Results Using Templates

Using the provided default-templates, you can create professional standard animations, even without having much expertise.





To do so, click Templates in the menu bar and select one of the templates. Then, enter the desired contents into the configuration dialogue. These contents are automatically inserted into the selected template and displayed as a ready-to-use product (including all animations and transitions).



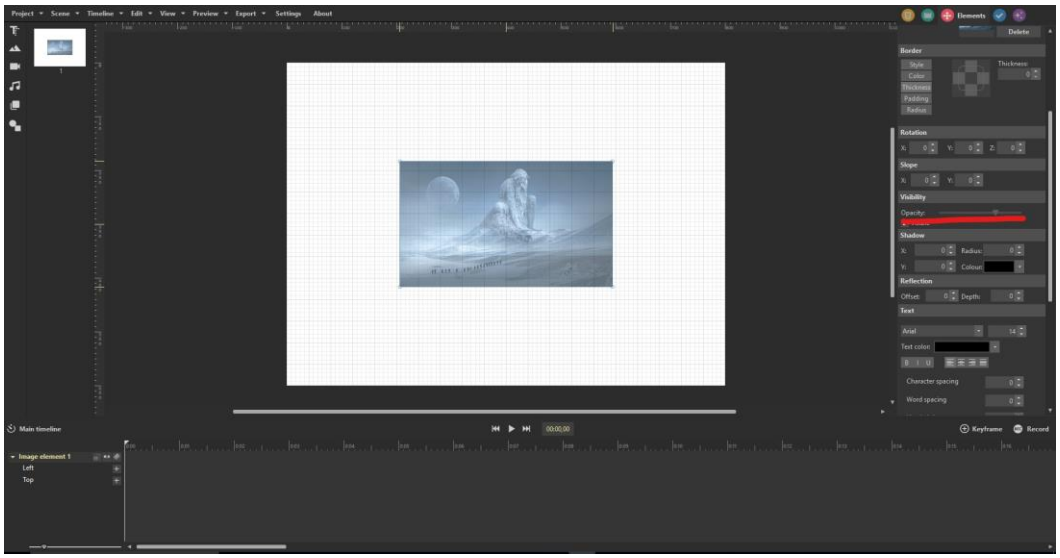
Examples

Basically, you are only limited by imagination when using WebAnimator. We would like to give you ideas and tips for your work with HTML5 animations with the help of a few different examples.

Fade-in and Fade-out

A popular animation effect is the fading in and out of objects. Using WebAnimator, this can be very easily implemented using the option *Opacity* in the *Properties* palette.

To do so, create an object on the stage. If you want to add other animations besides the fade-in and fade-out to the object, you can do it right away. However, you still have the option to do so later.

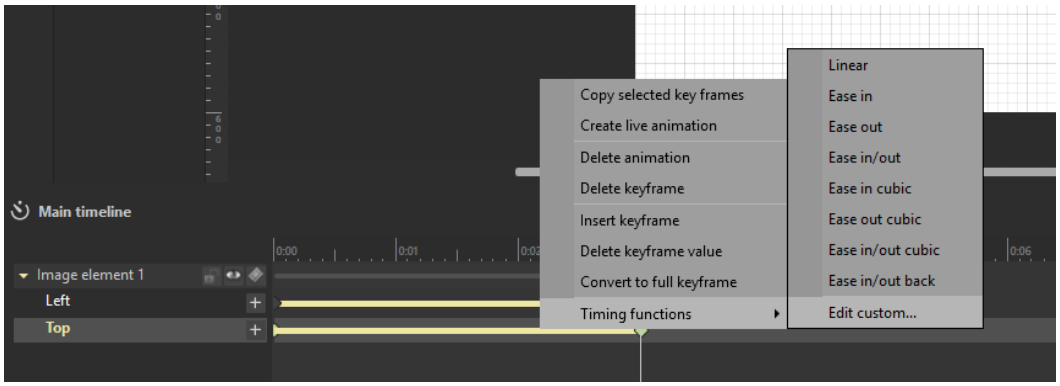


Create the keyframes for the animation as shown above and use the recording mode. For instance, the object will now be at full opacity at the beginning of the animation as the slider below Visibility is all the way to the right.

If you move the slider all the way to the left at the next keyframe, the object will fade out to invisible during the animation. In the same way, you can also have objects appear or become half transparent.

Timing function

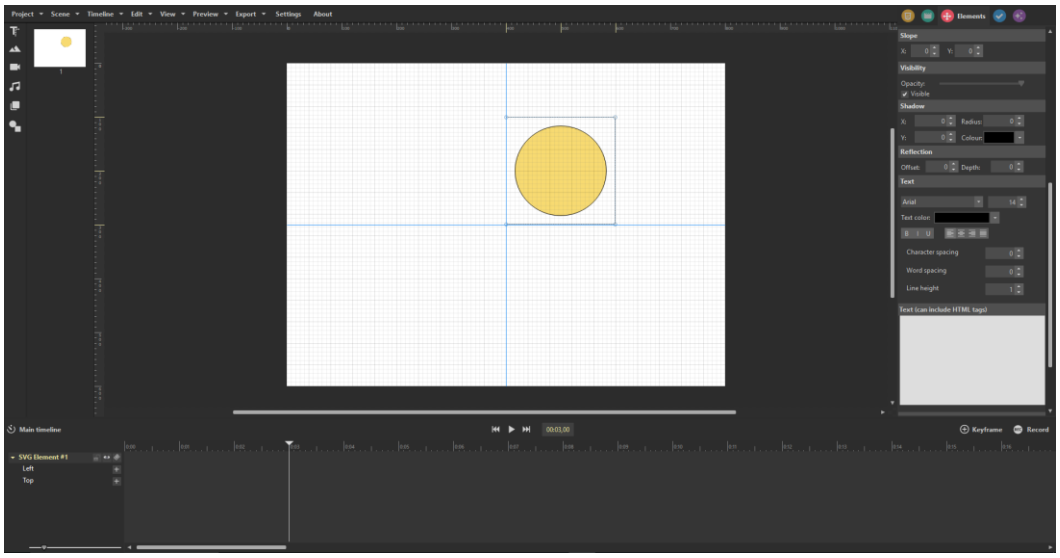
When you move an object in an animation, you have the option to add certain dynamics by setting the timing of the animation. To do so, select the animation in the timeline, then right-click the keyframe, and choose the option *Timing functions* from the context menu. Then, decide whether the movement is to be constant (*Linear*) or whether it should have a slight acceleration or deceleration effect (*Ease in / Ease out*).



Please do remember though that you really need to select the animation in the timeline and not just the keyframes. Furthermore, remember that this function can only be seen in the browser preview and not in the program itself.

Aligning Objects

WebAnimator gives you various options to align the object properly. For instance, a ruler function is provided by default at the top and left margin. The grid, which is also enabled by default, can help with aligning elements on the stage. You can however disable it via *View/Show grid lines*.

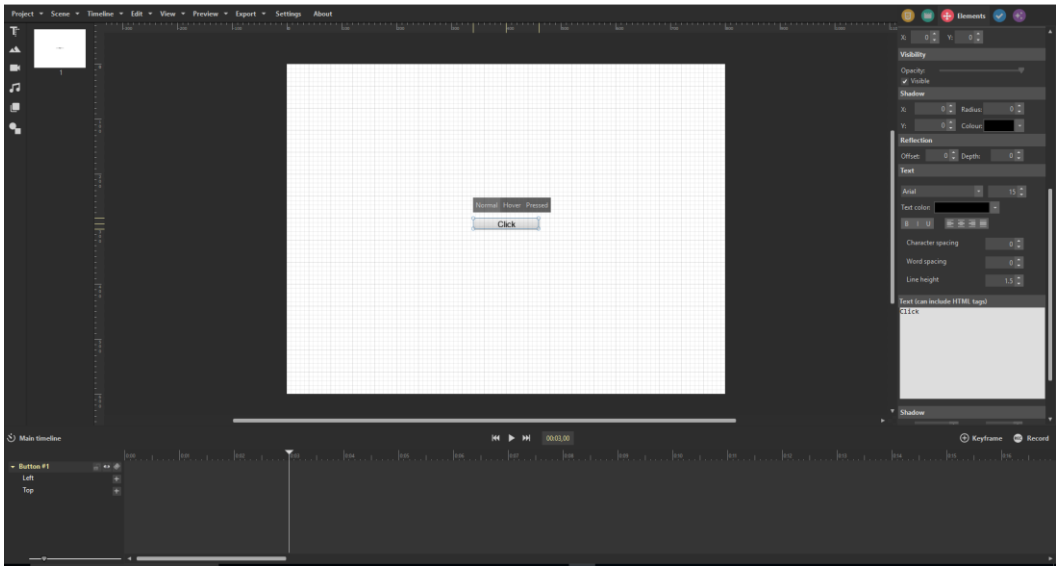


Another tool are the guides that automatically appear when you move an object vertically or horizontally close to the center while keeping the left mouse button pressed or when you move one object close to another one. These guides show you that you have at this moment aligned the object properly.

Buttons

A special type of object that you can use in WebAnimator are buttons. You can design individual buttons that can then serve as triggers for events, for instance.

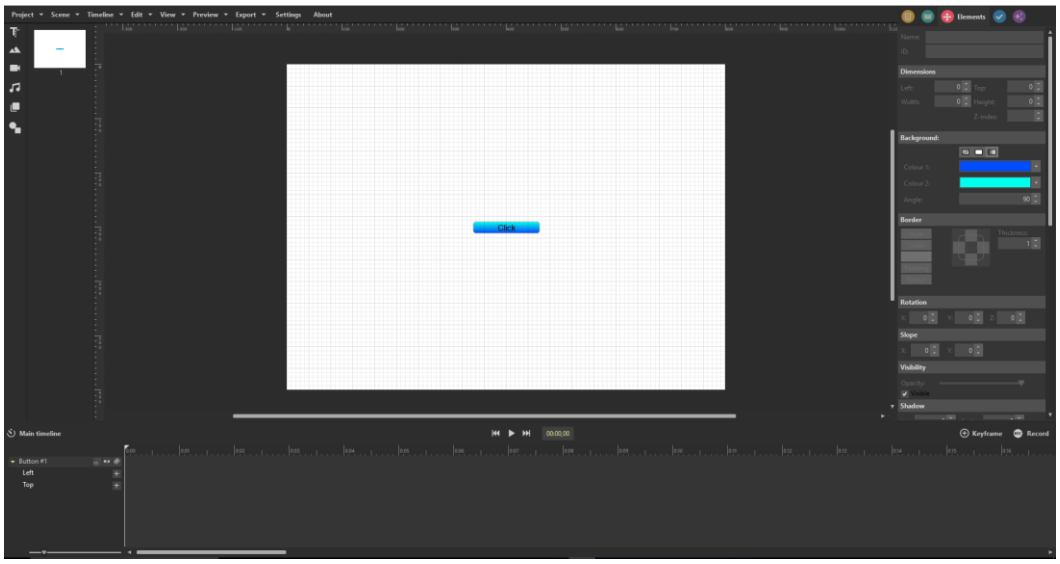
In order to create a button with this function, you simply drag, or click a button element onto the stage.



You can see that, besides the standard button which you can, of course, adapt according to your wishes, there are also two additional conditions included in this object. You can toggle between these with the three vertical buttons attached to the properties window. This gives you the opportunity to define different manifestations for the three possible options:

- Normal – standard option without any action by the user
- Hover – the user has moved the cursor on top of the object
- Pressed – the user has clicked the button

These manifestations may vary very subtly but also very drastically depending on the intended use.

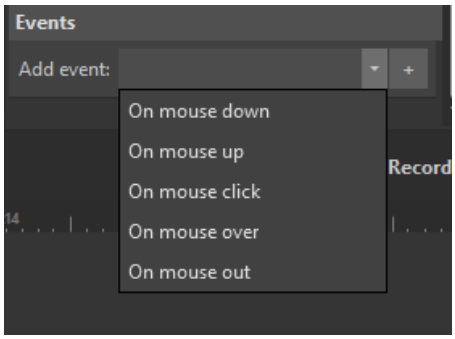


Creating Animations via Events

With WebAnimator, animations cannot just be carried out by changing the size, shape, color, and/or position of objects. You also have the option to design animations interactively and expand

them when the user manipulates the object. To do so, select an object on the stage and choose *Events* from the *Properties* palette.

This lets you define what should happen when a user performs certain actions within the animation. In doing so, you can affect the following conditions:

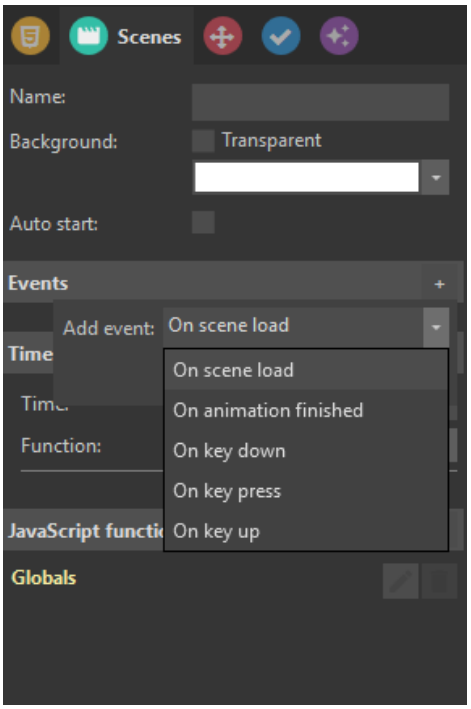


- *OnMouseDown*: As soon as the user presses the mouse button while the cursor is over the button, this event occurs.
- *OnMouseUp*: As soon as the user clicks the button by pressing and releasing the mouse button, this event occurs.
- *OnMouseClick*: While the user is clicking on the button, this event occurs.
- *OnMouseOver*: While the user moves the cursor over the button, this event occurs.
- *OnMouseOut*: When the user moves the cursor off the button, this event occurs.

Using individual events or a combination of these events, you can create complex, interactive animations. For instance, you can create a map of a route where individual parts are enlarged when the user moves over them with the mouse and where a link is opened when the user clicks the object.

However, events can also be assigned to scenes. The following conditions are possible:

- *OnSceneLoad*: As soon as the scene loads, this event occurs.
- *OnAnimationFinished*: After the animation is finished, this event occurs.
- *OnKeyDown*: When the user presses a key, this event occurs.
- *OnKeyPress*: When the user keeps a key pressed, this event occurs.
- *OnKeyUp*: When the user releases the button, this event occurs.



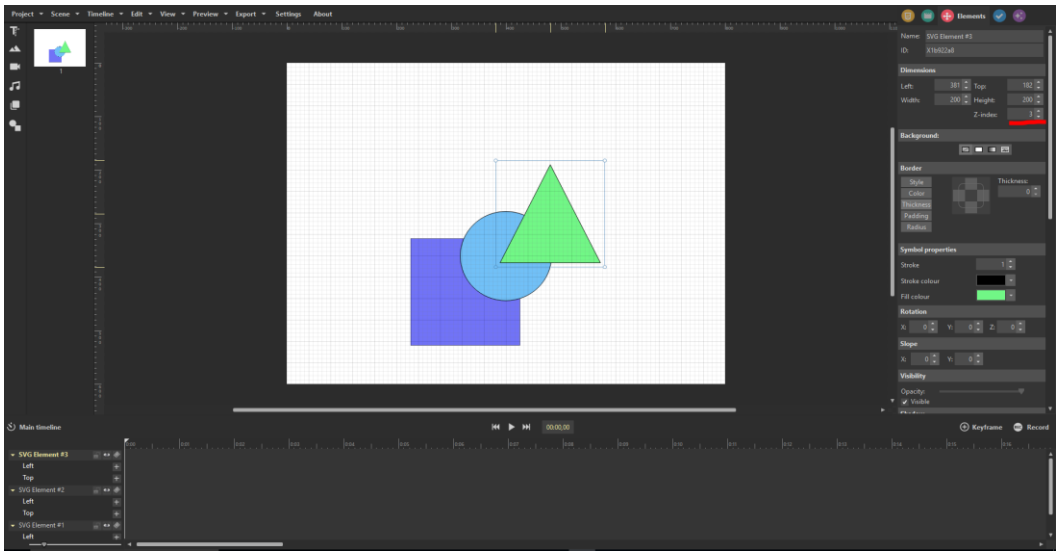
For instance, you can add a loop between scenes by specifying in *OnAnimationFinished* of a scene that the same scene is to be loaded again. The animation will then loop and replay again and again which can be used for banner ads, for instance.

Positioning Objects within the stage's space

In order to position objects in animations effectively, it is important to understand the different positioning options clearly.

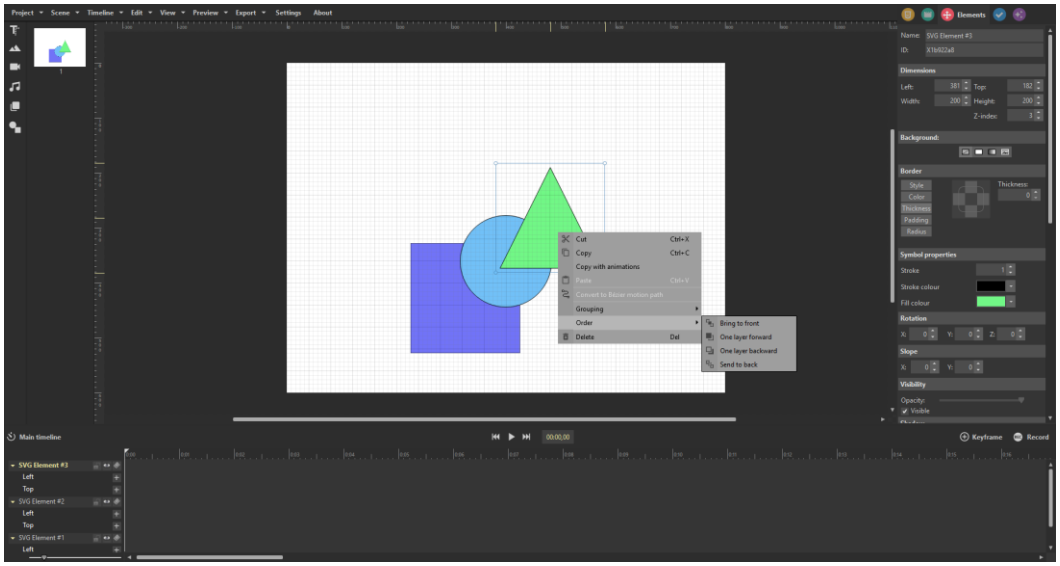
To start with, there are the simple spatial specifications of an object which can be seen in the details *Left* and *Top* in the timeline by default when you place an object. *Left* is the value located in the *Object Properties* palette in the *Dimensions* tab and which defines the distance to the left margin of the stage (*x-Position*). *Top* on the other hand is the value of the distance to the top margin (*y-Position*).

In the same tab, you will also find the value *z-index* which specifies how the elements are arranged on top of one another. By default, the elements are on top of one another in the sequence in which they are listed in the source text document; the element listed last is on top. By using the CSS property *z-index*, you can change this sequence.



Objects can therefore have three dimensions on the stage. The *x-position* and the *y-position* are displayed on the screen while the *z-index* determines which element lies on top of other elements. The higher the value of the *z-index* the closer to the top the element is.

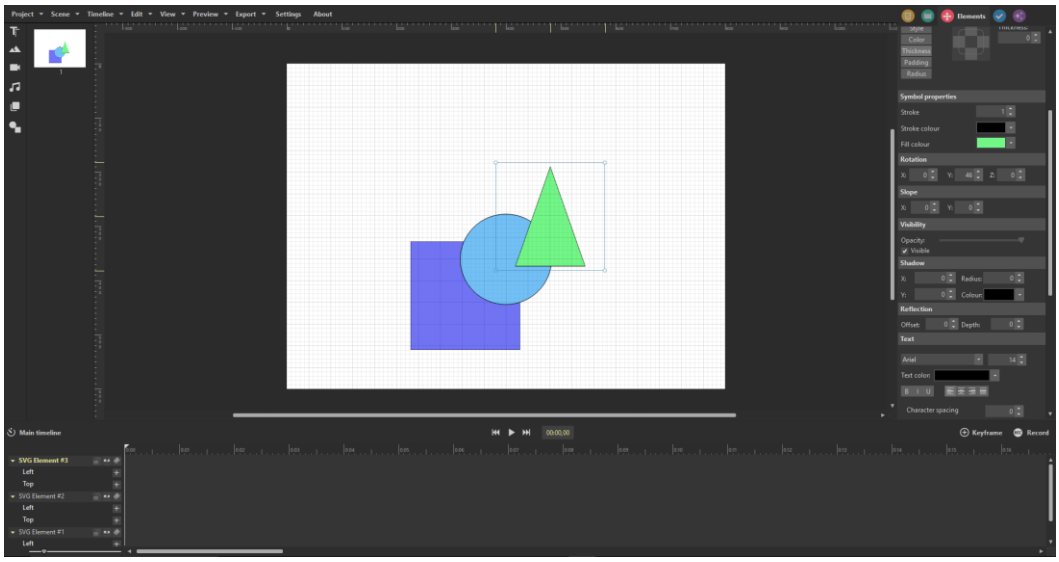
It is even easier to achieve this sort order if you select the relevant object and then select *Sorting* from the context menu. This gives you the option to move the respective element to the front, to the back, a layer forward, or a layer backward. You can then see this change in the details of the *z-index* in the *Properties* palette.



Last but not least, you have the option to grab the object description in the timeline and sort it via drag and drop.

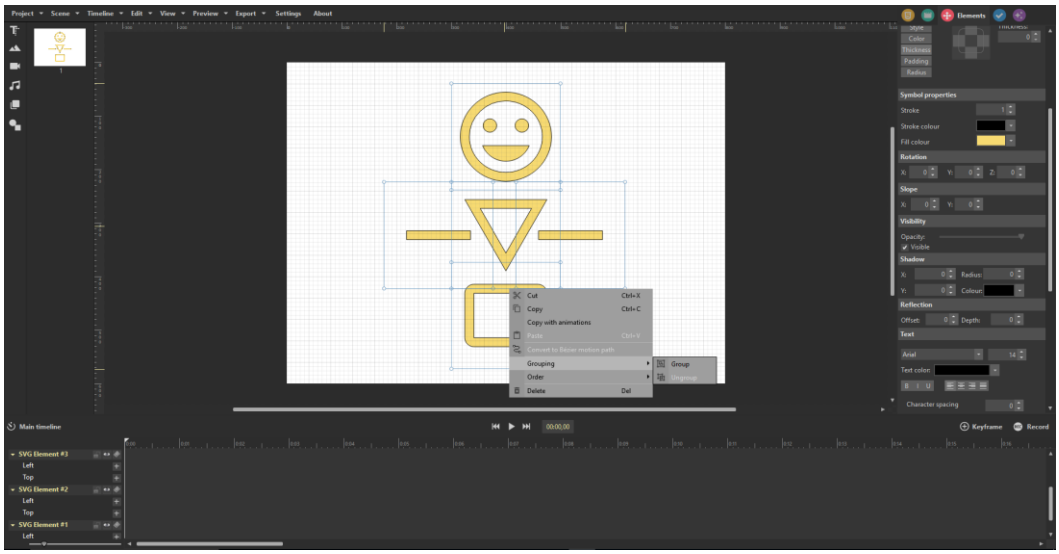
The *z-index* must not be confused with the rotation around the z-axis; the latter value is located in the tab *Element/Rotation*. When you place an object, such as a square, onto the stage, you must not envision it as two-dimensional but rather as a flat three-dimensional board.

Rotation around the x-axis will tilt the square to the back or the front; rotation around the y-axis will tilt it to the left or the right. Rotation around the z-axis will allow the object to revolve.



Grouping of Objects

You already know how to position and sort objects. In addition, you can also combine several objects into one. For instance, if you have created a complex object structure and want to place it at a different location on the stage or want to apply the same properties to all of the components of such a structure, you can combine them to one object by using *Grouping*.



To do so, select all the relevant objects by dragging a selection frame around them while keeping the left mouse pressed. Then click the right mouse button and select *Grouping/Group*.

Scenes and Interactions

As you can see, you can already create very complex animations with the tools described so far. In the following section, we want to go a step further and present professional tools and show you how to handle several scenes and timelines.

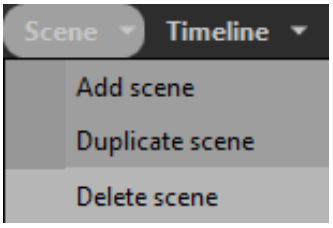
Using Scenes for Elaborate Designs

In principle, you can fit very complex animations into one single scene.

However, elaborate animations can be created and managed more easily if you use several scenes. This procedure is most easily compared to the individual slides of a presentation or the individual pages of a website.

Building Scenes

The actual creation of multiple scenes is simple. Just click *Scene* in the menu bar and then *Add scene* to create a new scene, or *Duplicate scene* to create a copy of the currently selected scene.



You can also delete scenes via the same menu. However, do remember that not all of the scenes can be deleted; at least one scene must remain.

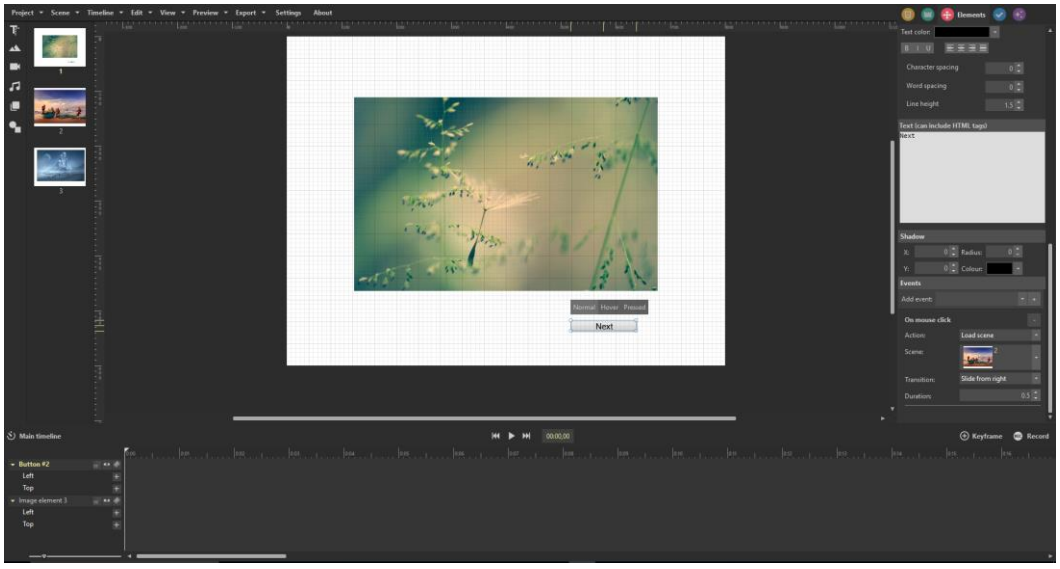
As shown above, you can change the size of a scene via the *Properties*. However, please remember that the size of the scene is a global value, i.e. you change the size of all the scenes when you adjust this parameter for one of them.

Inserting Transitions between Scenes

You have various options to go from one scene to the next, but you always have to use one of them because the animation will not be able to go to the next scene otherwise.

For instance, add a button to a corner of the first scene as shown above. Give the button the desired properties, such as size, design, and color until you are satisfied with it.

Then, select the button and choose the tab *Events* in the *Properties* palette. Here, you can specify what kind of transition should occur between one scene and the next.



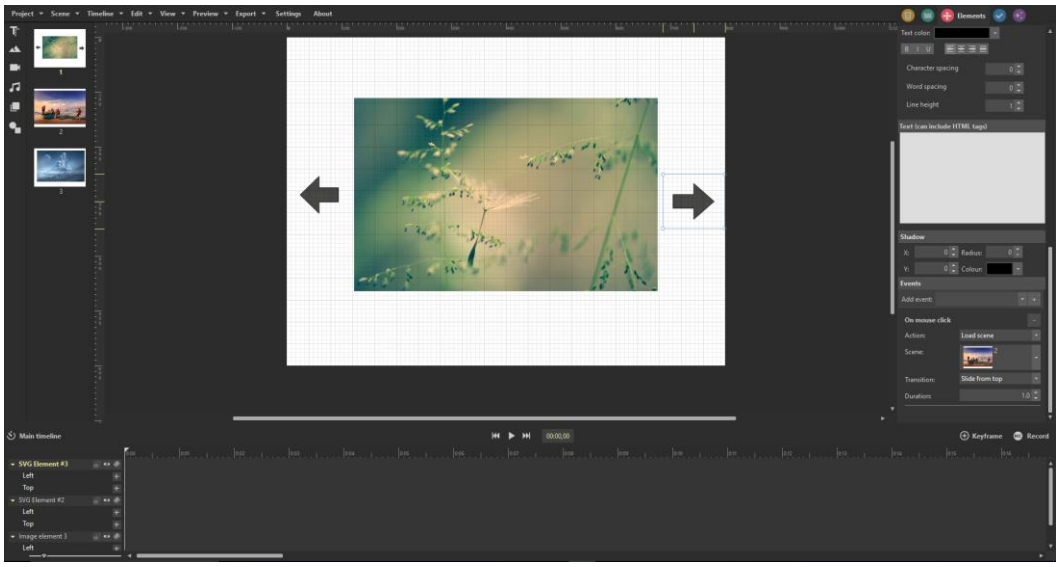
Since the new scene is not to be loaded until the button is pressed, select the option *OnMouseClick*. In order to have the next scene loaded when the button is clicked, under Scene

either select the scene to be loaded specifically from the list shown or select it based on how it is related to the current scene, such as Next scene.

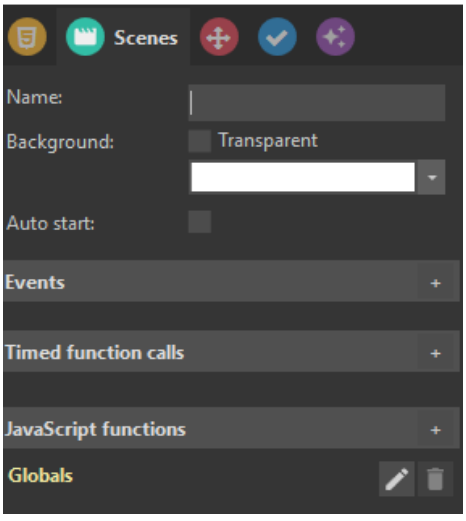
This procedure lets you freely define the sequence of the scenes which does not necessarily have to match the sequence of your project in WebAnimator.

In addition, you can decide what kind of transition you want between the two scenes. If you select *None*, the next scene will simply appear in the animation. You can also elegantly let the scene slide in from the left, right, above, or below in a similar way as in a presentation, or let it fade in dramatically. Below this option, you can define how long the effect should last.

In this way and using very simple tools, you can already create convenient sliders, such as product catalogues, virtual slide shows, etc. You can, of course, also define the image itself as the trigger for an action or, for instance, add an arrow image to the left and right of the image with which the user can easily go back and forth.



JavaScript editor

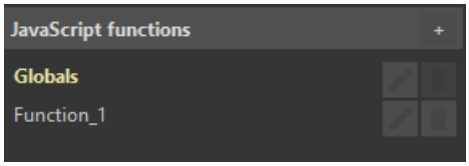


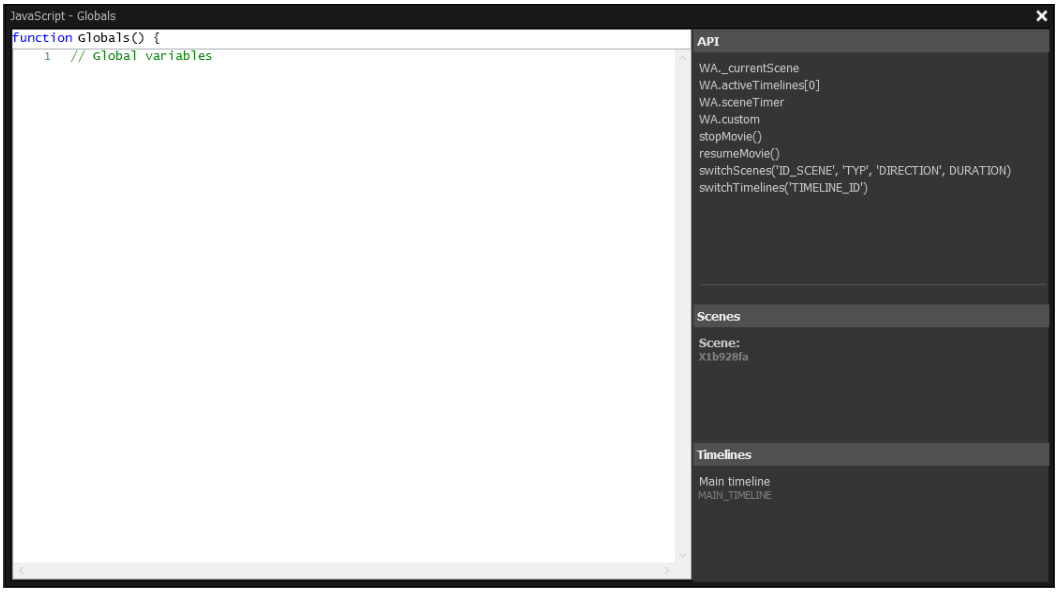
In prior sections, you have seen how you can, for instance, perform different actions in your animation by using scenes. However, you can also add your own JavaScript code to objects in order to execute even highly specialized functions.

To do so, first display the window of JavaScript functions by clicking Scenes in the tab bar and then JavaScript functions.

Then click the small plus icon in the JavaScript functions window to add a new function.

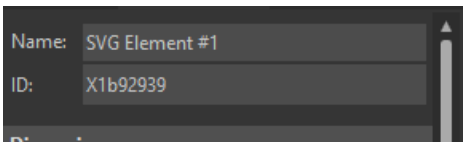
When you double click this function, the actual editor window opens.





Now you can start to write the JavaScript functions according to your ideas.

To the right, you will see various API functions that you can add to the editor via double-click. When you click one of these functions, a brief description of the function is displayed.



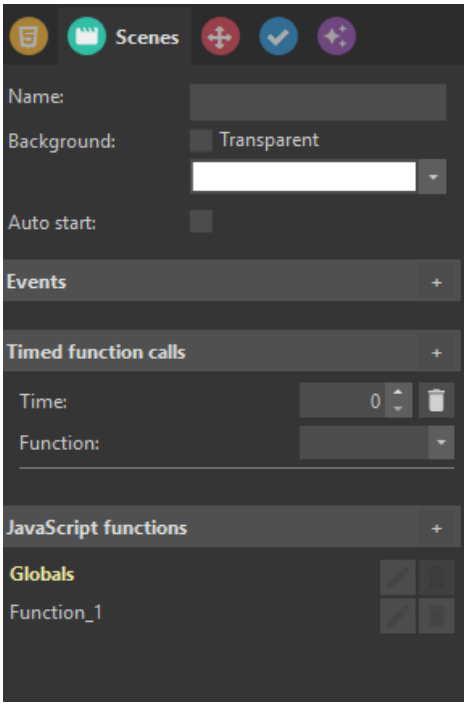
If you want to access an object from within your code, to manipulate it via jQuery, for instance, you will need the ID of the object.

The object ID is located in the *Object Properties* window in the *Element* tab under *ID*.

You may also change this ID there as you wish.

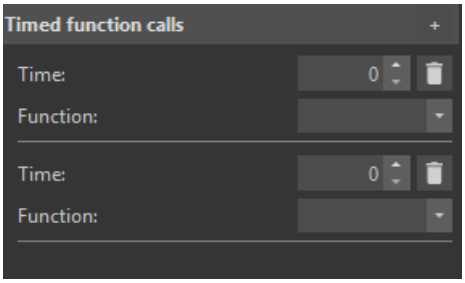
In order to assign your function to an object, to a button, for instance, select the function and then select *OnMouseClick* in the *Properties* window of the object in the *Events* tab or select one of the other possible event conditions. Select the option *Execute custom function* at the bottom and select the function you just created.

If you should require cross-function variables in your program code, use the entry *Globals* in the JavaScript functions manager. This entry serves as a storage location for global variables.



Scheduling your functions

All functions provided by WebAnimator in the form of events and JavaScript functions can be scheduled. You can configure the timing in the tab *Scene* of the *Properties* window.



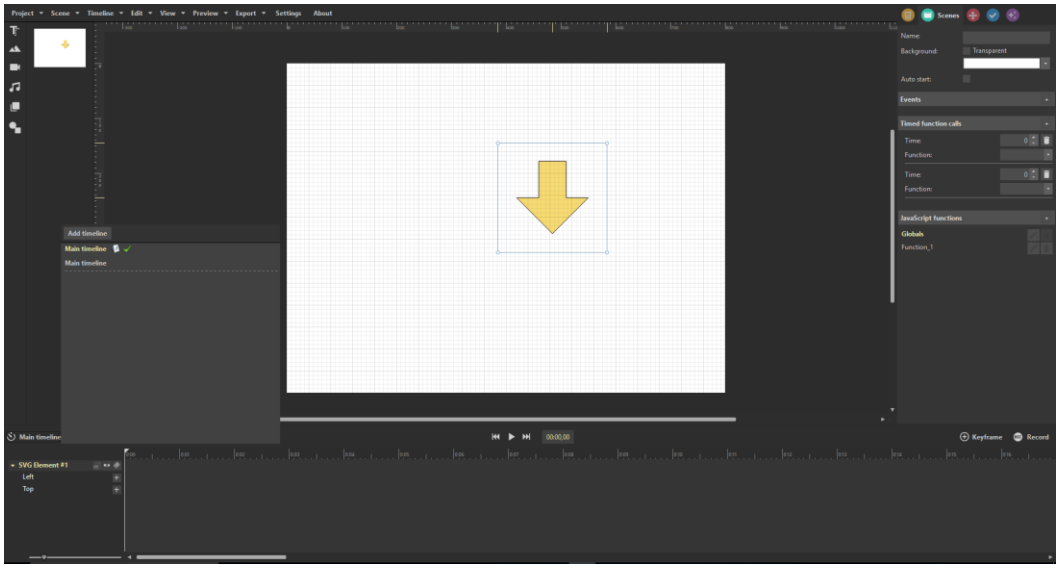
First click on "+" to create a function call. Define the point in time in which the function should be called and then the function itself. If you need a further function call you can create one by clicking on "+" again.

Multi-Timeline

You can also react to different events within a scene. For instance, it may be necessary to create two objects in one scene where the one object performs an animation while the other does not become active until the user moves the cursor over it.

You create objects that are only conditionally related to one another by using not one but several timelines. To do so, proceed as follows: Create the respective objects in a scene on the stage.

Then, create a second timeline by clicking on the button *Main timeline*. In a small popup window all timelines are listed. Use the button *Add timeline* to create a new one.

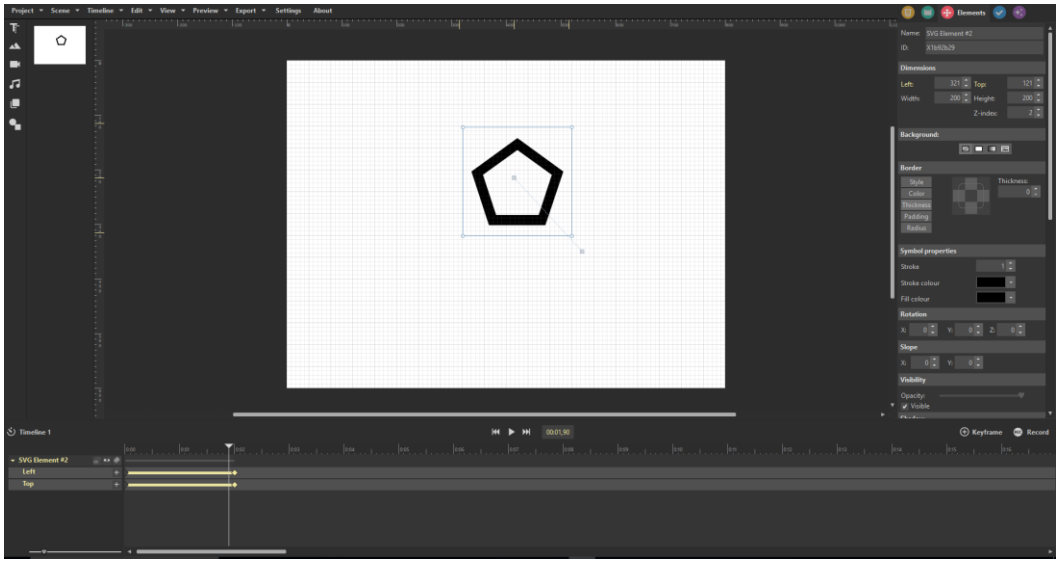


Click the small note icon once to customize the name of this timeline.

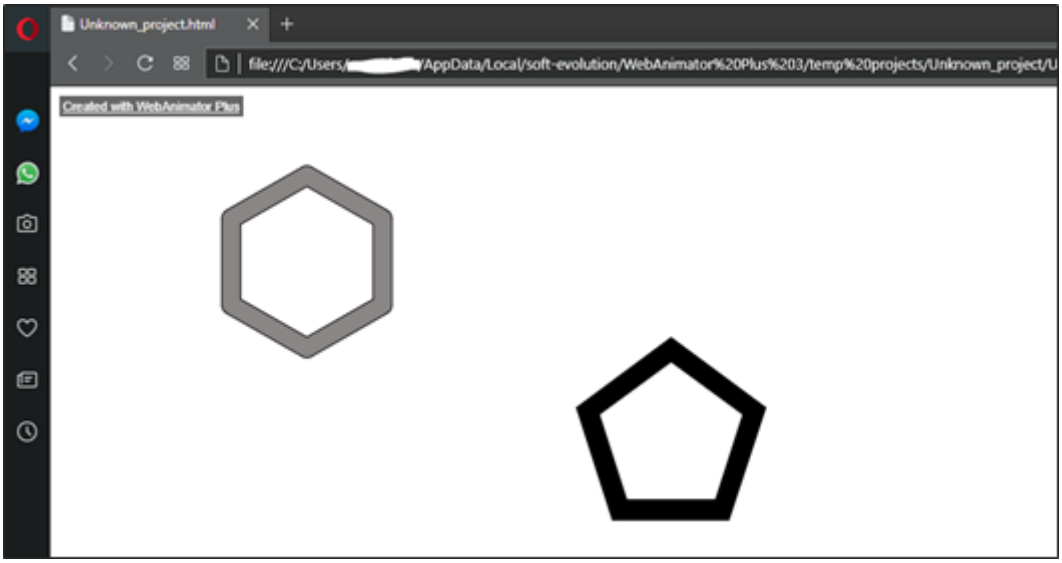
Now, you can switch between the timelines in the timeline overview with a click on the small green check icon next to the timeline.

However, nothing in your animation has changed so far. You can, however, animate the second object in the auxiliary timeline. This will not affect the actions in the main timeline.

Now, you have to assign an event to the object. To do so, select the object and choose the option *Events* from the *Properties* palette. Since you want to start the action in this example when the user moves the cursor over the object, go to the *OnMouseOver* section, click the action *Load timeline*, and select the auxiliary timeline.



Now, view all of it in the browser preview. The first animation runs right away while the second one does not start until you move the mouse over the object.



In this way, you can also add different interactive elements within a scene.